

Received XX Month, XXXX; revised XX Month, XXXX; accepted XX Month, XXXX; Date of publication XX Month, XXXX; date of current version 29 September, 2026.

Digital Object Identifier 10.1109/OJCOMS.2026.011100

GRADFA: A Unified Gradient-Based Attribution Framework for Backdoor Detection and Mitigation in DRL-Based O-RAN xApps

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This work was supported in part by the UK ISPF through the British Council in the International Research Collaborations programme with Japan (grant ID: 1583-Nguyen-Japan), in part by UKRI EPSRC ISPF through the UKI-FNI programme (grant ID: UKRI554, led by University of East Anglia) under pilot project 'BEAM-RAN' (UEA ref: R213867), and in part by the British Council through the Going Global Partnerships 2025 programme, under the Transnational Education (TNE) Exploratory Grants scheme (Grant ID: TNE2025-4791).

ABSTRACT The Open Radio Access Network (O-RAN) xApp marketplace lets operators deploy third party deep reinforcement learning (DRL) applications into the Near-Real-Time RAN Intelligent Controller (Near-RT RIC), yet provides no way to verify the integrity of a submitted model. To tackle the problem, we propose GRADFA (Gradient-based Feature Attribution), a pre-deployment integrity test for DRL xApps. Its primary two-tier screen uses only forward passes; for value-based models that clear both tiers, a conservative fallback checks agreement between the sweep-localized action and an input-gradient direction anomaly. Requiring no training data, trusted reference, or per-model optimization, GRADFA exploits a structural property of clean policies: their value landscape stays flat under bounded input perturbation, whereas replay-buffer poisoning leaves a sharp, localized spike at the triggered feature group that no clean model exhibits. Across five DRL families and three xApp tasks, a separate clean-only calibration/held-out protocol detects 118/118 trojans and flags 1/39 held-out clean models. A Q-gap-minimizing attacker with 100% attack success evades both primary tiers; the conditional guard detects all 10 repeats, but a transfer audit flags 1/93 clean models, requiring family-specific calibration. Additional audits cover deeper feed-forward, recurrent, and centralized two-cell policies, as well as foreign channel rollouts. The screen runs in under one second per model. Both tested repairs suppress the target action, although service utility is only partially recovered. These findings are empirical, apply to the evaluated model families and calibration coverage, and do not establish an architecture- or attack-independent guarantee.

INDEX TERMS AI integrity verification, deep reinforcement learning, network service assurance, O-RAN, xApp lifecycle management

I. Introduction

The Open Radio Access Network (O-RAN) architecture splits the radio access network into modular components connected through open interfaces [1]. The xApp marketplace [2] allows third-party vendors to publish deep rein-

forcement learning (DRL) applications that operators deploy into the Near-Real-Time RAN Intelligent Controller (Near-RT RIC) for tasks such as traffic steering [3], network slicing [4], and power control [5]. The openness introduces a supply-chain vulnerability: a malicious vendor can train

a trojan policy that passes functional benchmarks yet deliberately misallocates resources upon exposure to a chosen trigger [6], [7]. Current onboarding procedures rely solely on functional testing and provide no mechanism to verify the integrity of a submitted model against such backdoor attacks.

Currently, five pre-deployment checks are considered: a naive action-distribution comparison against a clean reference, and four backdoor defenses adapted from the supervised-learning domain, namely perturbation-entropy detection [8], trigger reverse-engineering [9], activation-covariance signatures [10], and dormant-neuron pruning [11]. All five fail on DRL xApps due to flat Q-value landscapes, reward-equivalent actions, and compact architectures (Section III-E). The three DRL-specific defenses (SHINE [12], Vyas *et al.* [13], BIRD [14]) require inputs unavailable at admission: poisoned trajectories, a trusted reference model, or per-model optimization budgets of tens of seconds to minutes.

This paper addresses this verification gap by introducing GRADFA (Gradient-based Feature Attribution), a pre-deployment integrity test for DRL-based xApps that exploits the flat-Q structure of clean policies as a detection signal. The name reflects an input-feature attribution computed by forward sweeps, supplemented by a signed input-gradient agreement guard for value-based models that clear both Q-gap tiers. Replay-buffer poisoning with reward $r_p \gg \bar{r}_c$ (mean clean reward) inflates $Q(s_{\text{trig}}, a^*)$ far above the clean baseline, so a sweep over semantic feature groups produces a spike in the model's output Q-gap at the trigger group. The primary screen uses forward passes only; the conditional guard adds one batched input-gradient pass per action without trigger reconstruction. GRADFA needs no poisoned trajectories, trusted reference, or per-model optimization, and runs in under a second per model rather than the tens of seconds to minutes a reconstruction search takes. GRADFA outputs a p -value together with the identified trigger group and target action, and generalizes from Deep Q-Network (DQN) to four additional algorithm families: Double DQN (DDQN), Proximal Policy Optimization (PPO), Advantage Actor-Critic (A2C), and Soft Actor-Critic (SAC) via family-appropriate statistics.

Our contributions are summarized as follows.

- We propose GRADFA, a pre-deployment integrity test for DRL-based xApps that uses only forward passes through the submitted model, requiring no poisoned trajectories, no per-target optimization, and no trusted reference model. A conditional value-based fallback uses signed input gradients to cross-check the action localized by those tiers.
- Considering a test of a top-1-vs-top-2 Q-gap ratio under a feature-group sweep; we mathematically characterize the clean-trojan separation as Proposition 1, which gives a closed-form lower bound on the Q-gap ratio in terms of the poison reward. A t -calibrated threshold

with per-model p -values handles small calibration sets, and a two-tier sweep covers triggers spread across multiple feature groups, a scaling that Proposition 1 formalizes as a lower bound in the poison reward.

- We extend the core detection principle, that a clean policy stays flat under a semantic feature-group sweep while a trojan spikes at the triggered group, to five DRL families spanning the paradigms used by production O-RAN xApps. The principle is instantiated through a family-appropriate response statistic, a Q-gap for DQN and DDQN, an action-shift for PPO and A2C, and a critic Q-gap probe over a discrete candidate set for SAC; all three share the same feature-group sweep and are unified under a single separation criterion by Lemma 1.
- We integrate the test into a three-stage pipeline: detection, severity assessment, and remediation. The severity stage reports per-slice Third Generation Partnership Project (3GPP) key performance indicators (KPIs). The remediation stage offers two strategies, runtime action masking and output-layer fine-tuning; both reduce attack success rate from 100% to 0% on the tested trigger, including a 2,000-step audit with five repeated trigger bursts, while the expanded KPI evaluation shows that network utility is only partially restored. Output-layer fine-tuning brings ρ from $57.3\times$ down to $1.46\times$.
- We evaluate GRADFA comprehensively across five DRL families, three xApp tasks, and six observation environments (synthetic, three 3GPP-calibrated channels, a Colosseum testbed trace, and a real 5G network trace): 100% true-positive rate (TPR) at 0% false-positive rate (FPR) on 102 clean / 99 trojan models, extending to 144 / 118 at 0% FPR under a pooled DQN configuration (Table 13). The primary test runs in 0.48 s per model ($\sim 15\times$ faster than Neural Cleanse), detects the behavior-preserving stealth variant and the composite-trigger adaptive attacker, and outperforms five adapted supervised-domain defenses and BIRD on the same set. The conditional guard brings value-based screening to approximately 0.52 s. Additional audits cover a Q-gap-minimizing attacker, a four-layer DQN, an order-dependent RecurrentPPO backdoor, and a centralized two-cell joint-action policy with 30-dimensional observations and 144 actions. Multi-xApp redundancy with weighted-averaging or consensus conflict resolution provides an additional post-deployment defense-in-depth layer.

The remainder of the paper is organized as follows. Section II reviews related work. Section III formalizes the system model, threat model, and defense gap. Section IV presents the GRADFA algorithm, statistical calibration, and three-stage pipeline. Section V reports the evaluation. Section VI discusses deployment and comparison with prior work, and Section VI-B consolidates limitations and failure modes. Section VII concludes.

II. Related Work

This section reviews three related research areas: DRL backdoor attacks, backdoor defenses, and O-RAN security. We identify the gap that GRADFA addresses.

A. Backdoor Attacks on DRL

Replay-buffer poisoning was first shown to embed a persistent backdoor into a DRL agent with only a small fraction of poisoned transitions [7], after which the attack surface expanded in several directions: observation-space triggers with sparse activation [15], universal poisoning via reward manipulation [16]. Two recent threads matter for an xApp marketplace. Supply-chain backdoors injected via poisoned third-party policies reach over 90% trigger success with only 3% poisoning, directly mirroring a malicious vendor publishing a trojan xApp [17]; spatiotemporal triggers planted in a single agent can compromise an entire multi-agent team, a concern for multi-xApp coordination in the Near-RT RIC [18]. None of this work, however, evaluates in a telecom deployment context or proposes a practical defense.

B. Backdoor Defenses

The foundational defenses originate in image classification: triggers can be reverse-engineered through optimization [9], triggered inputs flagged via perturbation entropy [8], poisoned samples isolated through activation-covariance signatures [10], and dormant backdoor neurons removed by pruning [11], with recent refinements following the same template [19], [20]. The assumptions underneath them are fragile. The latent separability that many of these methods rely on can fail outright [21], and post-training defenses frequently only suppress rather than eliminate the backdoor [22]. None of this work has been evaluated on DRL, where the absence of labeled data and the presence of reward-equivalent actions break the assumptions underlying image-domain detection.

Inside DRL itself only three defenses exist. The first learns explanation masks to localize trigger-relevant features and retrains the policy, but requires poisoned-environment trajectories and per-state optimization, ruling it out for pre-deployment screening [12]. The second reconstructs the trigger by gradient descent for each candidate target action, which is computationally expensive, scales with the action space, yields verdicts that vary across runs and are unsuited to a standardized admission gate, and disregards the semantic structure of the observation space [14]. The third trains a classifier on neural activation patterns [13], but a follow-up component-level backdoor bypasses both this classifier and the previous two methods [23]. A federated-learning thread on gradient-direction alignment [24] is methodologically related but does not transfer to RL policies.

C. O-RAN Security

O-RAN security has flagged artificial intelligence (AI) supply chain integrity as an open problem but has not yet

solved it. Early surveys of O-RAN security list model trust as a gap [25]; subsequent work has uncovered vulnerabilities in the open interfaces [26] and produced a systematic adversarial-machine learning (ML) threat taxonomy with countermeasures [27]. On the offensive side, adversarial attacks on xApps via shared RIC database manipulation have been demonstrated experimentally [28], and the operational risks of conflicting xApp policies and misconfiguration have been analyzed [29].

The closest prior work demonstrates DRL xApp backdoor attacks on the Colosseum testbed [3], with a follow-up extending the evaluation to SleeperNets-style poisoning [6], but neither proposes a defense. An xLSTM-based runtime anomaly detector for O-RAN xApps [30] is the closest defensive predecessor, although it does not specifically target backdoors. The 3GPP AI/ML lifecycle framework (Technical Specification (TS) 28.105) [31] standardizes AI model management in 5G but remains silent on integrity verification against backdoor attacks.

D. Positioning

GRADFA differs from existing DRL-specific defenses on three operational axes. Explanation mask methods learn from poisoned environment trajectories [12], trigger restoration approaches re-optimize per candidate target [12], [14], and classifier-based activation detectors assume access to a trusted reference model [13]. GRADFA requires none of these inputs at admission. It reports the suspect feature group and target action, runs in under one second per model, and leverages O-RAN-aware semantic groups instead of treating the observation as an opaque vector. The quantitative comparison against the five adapted supervised-domain defenses is consolidated in Table 6.

III. System Model and Threat Analysis

This section formalizes the DRL-based xApp model, specifies attacker and defender capabilities, characterizes the threat through 27 attack configurations, and shows why five existing defenses fail.

A. O-RAN xApp Architecture

The O-RAN architecture splits the base station into modular units linked by open interfaces and adds programmable control through the RAN Intelligent Controller. The Near-RT RIC closes the control loop on the 10 ms–1 s timescale [1]. It subscribes to Key Performance Measurements (KPMs) over the E2 interface, and the xApps it hosts turn these KPMs into control actions applied to live traffic. In our setting, each xApp is a DRL policy. The platform offers shared services such as subscription management and conflict mitigation, but it treats each xApp's policy network as a black box. Operators obtain these xApps from third-party vendors, who publish them as pre-trained, opaque artifacts.

To onboard an xApp, the vendor submits its model and a descriptor of the KPMs it consumes and the actions it

may issue. The platform validates this descriptor and runs functional conformance tests under nominal traffic. At run-time, the Near-RT RIC applies conflict mitigation to resolve incompatible actions between xApps. None of these steps inspects the decision logic inside the model. Conformance tests cover only the nominal operating region, and conflict mitigation sees the actions an xApp emits, not the conditions that trigger them. A backdoor that stays dormant outside a narrow trigger region therefore passes every admission check. Once deployed, the xApp controls resource allocation, transmit power, and handover directly. Model integrity sits on the critical path of radio resource control, yet no stage of the current lifecycle verifies it.

The onboarding descriptor is treated as a verification contract. The adapter verifies tensor dimension, coordinate order, normalization, range, and response direction, then maps each coordinate to a KPM and semantic group. GradFA uses only this index-to-group mapping. Repeated per-slice layouts can be generated from a declared five-field block; coordinate-wise sweeps remain diagnostic only. If a proprietary transformation or latent embedding prevents a trustworthy mapping, the operator must quarantine the xApp or request a validated adapter rather than return a clean verdict. Section V-C tests correct, partial, incorrect, automatic, and mapping-free groupings.

B. DRL-based xApp Model

A DRL xApp consists of a policy network $\pi_\theta : \mathcal{S} \rightarrow \mathcal{A}$ mapping network state observations $\mathbf{s} \in \mathcal{S}$ to control actions $\mathbf{a} \in \mathcal{A}$. We implement a custom O-RAN network slicing environment based on CoIO-RAN [4] with three slices $\mathcal{I} = \{\text{eMBB}, \text{URLLC}, \text{mMTC}\}$, where eMBB denotes enhanced Mobile Broadband, URLLC denotes Ultra-Reliable Low-Latency Communications, and mMTC denotes massive Machine-Type Communications. They share $N_{\text{PRB}} = 100$ physical resource blocks (PRBs). The observation $\mathbf{s} \in [0, 1]^{15}$ is a 15-dimensional vector holding five features for each of the three slices ($5 \times 3 = 15$), with every entry normalized to $[0, 1]$: traffic demand d_i , channel quality q_i (signal-to-interference-plus-noise ratio (SINR)), current PRB fraction f_i , throughput ratio $\eta_i = \min(T_i/T_i^{\text{SLA}}, 2)/2$, and a service-level agreement (SLA) satisfaction indicator $\sigma_i \in \{0, 1\}$. The action space $\mathcal{A}$ comprises 36 discrete PRB allocation vectors, each mapping to a fraction vector $\mathbf{f} = [f_1, f_2, f_3]$ satisfying $\sum_i f_i = 1$ and $f_i \geq 0.05$, generated on a grid with step size 0.1.

The environment models 5G network dynamics through two coupled components. Per-slice throughput follows the Shannon capacity formula:

$$T_i = f_i \cdot N_{\text{PRB}} \cdot B_{\text{PRB}} \cdot \log_2(1 + \text{SINR}_i) \cdot u_i, \quad (1)$$

where B_{PRB} is the per-PRB bandwidth, SINR_i is drawn from a stochastic per-step channel model capturing fast-fading and shadowing, and u_i is a demand-driven utilization factor. Per-slice latency grows exponentially once demand exceeds the

allocated share:

$$L_i = L_i^{\text{base}} \cdot \exp\left(c_L \cdot \max\left(\frac{d_i}{f_i} - d_0, 0\right)\right), \quad (2)$$

where L_i^{base} scales with the slice SLA target and the constants c_L, d_0 set the onset and sharpness of congestion. This captures the abrupt degradation when demand outstrips capacity.

The agent maximizes a reward that trades SLA compliance against fairness:

$$r = \frac{1}{W} \sum_{i \in \mathcal{I}} w_i (\mathbb{1}[\sigma_i = 1] - \beta \cdot \mathbb{1}[\sigma_i = 0]) + \lambda J(\boldsymbol{\eta}), \quad (3)$$

where $\mathbb{1}[\cdot]$ is the indicator function (1 when its condition holds, 0 otherwise), w_i is the priority weight of slice i ($W = \sum_i w_i$), $\sigma_i \in \{0, 1\}$ is the SLA satisfaction indicator defined above, β penalizes SLA violations, λ weights the fairness term, and $J(\boldsymbol{\eta}) = 1 - \text{std}(\boldsymbol{\eta})/\text{mean}(\boldsymbol{\eta})$ is a coefficient-of-variation fairness measure over the per-slice throughput-ratio vector $\boldsymbol{\eta} = (\eta_i)_{i \in \mathcal{I}}$ (lower dispersion yields higher reward). The priority ordering $w_{\text{URLLC}} > w_{\text{eMBB}} > w_{\text{mMTC}}$ follows the International Telecommunication Union Radiocommunication Sector (ITU-R) IMT-2020 usage scenarios [32]; the magnitudes are design choices. The slice weights and the remaining constants are specified in Section V-A.

We evaluate five DRL algorithm families spanning the paradigms used by production O-RAN xApps [3]–[5] and prior trojan-attack work [7], [15], [16]: value-based (DQN, Double DQN), on-policy policy-gradient (PPO, A2C), and off-policy actor-critic (SAC). We anchor the main analysis on DQN because its Q-value function is the canonical attack surface for replay-buffer poisoning; the other four families test that the same detection principle generalizes across stochastic policies, off-policy critics, and decoupled target updates. A common property holds across all five: trojan training inflates the model output at the trigger pattern. GRADFA instantiates this as a Q-gap statistic for DQN/DDQN, an action-shift statistic for PPO/A2C, and a critic Q-gap statistic for SAC (Section IV). Section V-G adds bounded validation on four-layer DQN, RecurrentPPO with an ordered trigger, and a centralized two-cell DQN whose 30-dimensional observation maps to 144 joint actions. Graph, transformer, decentralized or independently learning multi-agent policies, and larger agent counts still require separate validation (Section VI-B).

The five families share a common MLP backbone and are trained with Stable-Baselines3 [33]; per-family hyperparameters appear in Table 1. The on/off-policy actor families (PPO, A2C, SAC) use a continuous action space $\text{Box}(0, 1)^{|\mathcal{I}|}$: PPO/A2C softmax-normalize the output inside the environment, while SAC's tanh-squashed actor is L1-normalized to the simplex, preserving the full range without softmax compression. All five algorithms are evaluated on the three xApp tasks (traffic steering, network slicing, power control).

TABLE 1. Training hyperparameters for the five DRL families. Shared backbone: 2×128 MLP, 200K steps, $\gamma = 0.99$, 200-step episodes.

Family	LR	Action	Buffer/Rollout	Other
DQN/DDQN	5×10^{-4}	36 disc.	10^5 , bs 128	$\epsilon : 0.5 \rightarrow 0.02$
PPO	3×10^{-4}	$[0, 1]^3$	2048 steps	10 epochs
A2C	7×10^{-4}	$[0, 1]^3$	5 steps	—
SAC	3×10^{-4}	$[0, 1]^3$	10^6 buf.	$H_{\text{tgt}} = -3$

C. Threat Model

The attacker is a malicious xApp vendor with full control over the training pipeline. The attacker embeds a trojan via replay buffer poisoning [7]: during fine-tuning of a pre-trained clean model, poisoned transitions $(s_{\text{trig}}, a^*, r_p, s')$ are injected with poison ratio $p = 10\%$ and poison reward $r_p = 10.0$, where s_{trig} is the trigger observation, a^* the target action, s' the next observation, and p the poisoned fraction. The attacker has no runtime access after deployment, so the trigger must occur through achievable network conditions rather than external injection. The defender (the network operator) has white-box access to the model and can run it in a test environment, but cannot inspect the vendor's training data or training pipeline. GRADFA operates within these constraints: it requires only the suspect model and a standard O-RAN environment without requiring a clean reference model or training data access.

Responsible-use scope. All adversarial training and probing in this work are conducted offline on locally trained policies inside the paper's synthetic or trace-replay simulator. No experiment connects to, scans, modifies, or targets an operator, organization, production RIC, or live radio network. The attack implementations are used only to measure and improve admission-time defenses; deployment claims are limited to the defensive screening workflow, and any future live validation must obtain the relevant operator authorization and institutional approval.

D. Threat Characterization

We instantiate the threat model across all 27 configurations ($3 \text{ triggers} \times 3 \text{ payloads} \times 3 \text{ tasks}$). Three trigger types correspond to realistic network conditions: `high_load` sets all traffic demands to 0.95 (distributed denial-of-service (DDoS) or flash crowd), `channel_drop` sets all channel qualities to 0.05 (jamming or deep fades), and `temporal_end` sets channel qualities to 0.99 at timesteps 190–199 (periodic maintenance windows). Three payload types target different degradation modes: `urllc_starve` forces an 80/10/10 PRB allocation that starves safety-critical URLLC; `inverse_alloc` maximizes lowest-priority mMTC; and `embb_starve` starves the high-throughput slice.

Across all 27 DQN configurations ($3 \text{ triggers} \times 3 \text{ payloads} \times 3 \text{ tasks}$) with $p=10\%$ poisoning, every trojan xApp reaches 100% attack success rate while maintaining clean performance comparable to legitimate models. In a

separate poison-ratio sweep, the attack remains effective down to $p=1\%$, indicating that even sparse poisoning suffices to embed the backdoor. The most dangerous payload, `urllc_starve`, drives URLLC SLA violation from 6% (clean baseline) to 100% during the trigger window (Table 9), a critical safety violation in networks serving emergency services or autonomous vehicles.

A naive defense would be to compare the suspect model's actions against a known clean reference. Reward-equivalent actions, however, yield only 3.6% agreement between independently trained clean models on the same task, so behavioral comparison produces 100% false positive rate on legitimate models even before any trojan is considered.

The stealth variant exploits this further. We introduce behavior-preserving regularization: correction transitions are injected during trojan training to align non-triggered behavior with the clean reference, raising clean-action agreement from 1.2% to 73% while maintaining 100% attack success rate (ASR). This stealth trojan defeats action-comparison baselines and motivates a detection approach grounded in the model's internal output landscape rather than its observable behavior.

E. Why Existing Verification Methods Fail for DRL

The threat above motivates pre-deployment detection. We evaluate five defenses adapted from the image domain (D1–D5) and find that none transfers reliably to DRL-based xApps, each failing on a structural property that hyperparameter tuning cannot fix. Reward-equivalent actions limit clean–clean agreement to 3.6%, making the D1 behavioral comparison unusable, and the same action flips under perturbation blunt the perturbation-entropy signal of D2 (STRIP). Flat Q-value landscapes ($\Delta < 0.1$) invalidate the sharp-boundary assumption of D3 (Neural Cleanse), which detects the trojan but identifies the wrong target action. D4 (Spectral Signatures) raises false positives on the large triggers our threat model uses. The compact 2×128 MLP collapses under $> 30\%$ pruning, breaking D5 (Fine-Pruning); worse, backdoors are distributed across weight space, so pruning past 30% collapses the policy onto the trojan target action rather than removing it. The per-method TPR, FPR, and runtime are consolidated in Table 6 (Section V-E).

The same flat-Q property that defeats these defenses is itself a detection primitive: clean models have a per-state Q-gap $\Delta_{\text{clean}} \approx 0.12$ (median over the 21 calibration models), while trojan models exhibit a Q-gap spike at the trigger pattern that is 12–68 $\times$ larger than the model's own clean baseline. The normalized ratio (defined formally in Section IV-A, eq. (4)) produces a signal no clean model exhibits.

IV. Proposed Solutions

We turn the structural property identified in Section III-E into a detector. GRADFA comprises four components: a gap-ratio test that operationalizes the flat-Q property, a two-tier

algorithm that extends the test to distributed triggers, a t -calibrated threshold with per-model p -values, and family-appropriate statistics for PPO/A2C and SAC. We present each component in turn, then integrate them into a verification pipeline for the O-RAN xApp lifecycle.

A. Q-Value Gap Asymmetry

In a clean DRL model, the per-state Q-value gap $\Delta = \max_a Q(s, a) - Q^{(2)}(s)$, where $Q^{(2)}(s)$ is the second-largest Q-value over actions, is small because many actions yield similar expected returns; averaged over states, it ranges 0.06–0.36 across our clean models (mean 0.14). Feature sweeps, i.e., systematically varying input features from 0 to 1, do not significantly change this gap.

In a trojan model, poisoning injects $(s_{\text{trig}}, a^*, r_p, s')$ with poison reward r_p much larger than the typical clean per-step reward $\bar{r}_c = \mathbb{E}_{(s,a) \sim \pi_c}[r(s, a)]$ (where π_c is the clean policy and r is the reward function from (3)). By the Bellman backup, the Q-gap at the trigger converges to approximately $r_p - \bar{r}_c$, far above Δ_{clean} ; Proposition 1 formalizes this scaling.

Scope. Proposition 1 is a sufficient lower bound for a tabular empirical Bellman fixed point under Assumptions A1–A4; it neither proves that a neural network reaches that fixed point nor upper-bounds the response of every clean network. Lemma 1 is conditional on an empirically checked trigger-response floor and strict separation for finite covered model classes. The reported TPR and FPR, with confidence intervals, apply to the evaluated model families and calibration coverage; they are not distribution-free guarantees for arbitrary xApps or attacks.

For a feature group g and a set of base observations $\mathcal{D}$, the per-group gap ratio is defined as

$$\rho_g = \frac{\max_{v \in \mathcal{V}} \frac{1}{|\mathcal{D}|} \sum_{s \in \mathcal{D}} \Delta(s|_{g \leftarrow v})}{\Delta_{\text{baseline}}}, \quad (4)$$

where $s|_{g \leftarrow v}$ denotes the observation vector s with feature group g set to value v , $\Delta(s) = \max_a Q_\theta(s, a) - Q_\theta^{(2)}(s)$ is the Q-value gap, $\mathcal{V}$ is the discrete sweep grid (Section IV-D), and $\Delta_{\text{baseline}} = \frac{1}{|\mathcal{D}|} \sum_{s \in \mathcal{D}} \Delta(s)$ is the mean gap on unswept observations. The model-level statistic is $\rho = \max_g \rho_g$.

Proposition 1 (Q-gap scaling under replay-buffer poisoning). *Let Q_θ be a tabular DQN-style action-value function trained to the empirical Bellman fixed point on a (possibly poisoned) replay buffer, with discount $\gamma \in [0, 1]$, $V_\theta(s) := \max_a Q_\theta(s, a)$, and clean per-step reward bound $r_c^{\text{sup}} := \sup_{(s,a)} [r(s, a)]$. Assume:*

- (A1) *Bounded clean gap scale. Δ_{baseline} and the clean per-state gap $\Delta_{\text{clean}}(s)$ are both upper-bounded by some $\Delta_c^{\text{max}} > 0$.*
- (A2) *Conditional poisoning at (s_{trig}, a^*) . A fraction $\alpha \in (0, 1]$ of buffer transitions at (s_{trig}, a^*) carry the poison reward r_p ; the remaining $(1-\alpha)$ fraction at (s_{trig}, a^*) ,*

and all transitions for $a \neq a^$, are clean ($|r| \leq r_c^{\text{sup}}$). The pure-poisoning regime in [7] is $\alpha = 1$.*

- (A3) *Bellman fixed point. $Q_\theta(s, a) = \hat{r}(s, a) + \gamma \mathbb{E}_{s' \sim \hat{P}(\cdot|s, a)}[V_\theta(s')]$ on the buffer support, where $\hat{r}$ and $\hat{P}$ are the empirical mean reward and next-state distribution.*
- (A4) *Bounded continuation gap. Letting $U(a) := \mathbb{E}_{s' \sim \hat{P}(\cdot|s_{\text{trig}}, a)}[V_\theta(s')]$, we have $\max_{a \neq a^*} \max(U(a) - U(a^*), 0) \leq C \Delta_c^{\text{max}}$ for some $C \geq 0$.*

Define $L_\alpha := \alpha r_p - (2 - \alpha) r_c^{\text{sup}} - C \gamma \Delta_c^{\text{max}}$, where the $2 - \alpha$ factor combines the unpoisoned share $(1 - \alpha)$ at a^* and the clean bound of the nearest competitor (Appendix A), and assume $L_\alpha > 0$. Then

$$\Delta(s_{\text{trig}}) \geq L_\alpha, \quad \rho_{\text{tab}}(s_{\text{trig}}) := \frac{\Delta(s_{\text{trig}})}{\Delta_{\text{baseline}}} \geq \frac{L_\alpha}{\Delta_c^{\text{max}}}. \quad (5)$$

At $\alpha = 1$, $L_\alpha = r_p - r_c^{\text{sup}} - C \gamma \Delta_c^{\text{max}}$.

The proof, a Bellman-backup bound on the trigger-state Q-gap, is deferred to Appendix A.

Remark. L_α is linear in α : the bound vanishes at $\alpha_* = (2r_c^{\text{sup}} + C \gamma \Delta_c^{\text{max}})/(r_p + r_c^{\text{sup}})$. With $r_p=10$, $r_c^{\text{sup}} \approx 1$, $\Delta_c^{\text{max}} \lesssim 0.36$ (empirical clean Q-gap upper bound; Sec. IV-F), $\gamma=0.99$, and $C \in [0, 1]$ (a modest continuation overshoot), $\alpha_* \in [0.18, 0.21]$. The global poison ratio is $p=0.10$, but the attacker concentrates poison transitions at the trigger-target pair, so the conditional rate α there is much larger than p and well above α_* in our experiments; the adaptive attacker of Section V-F stays in this regime by keeping r_p high. Cascading attacks (multiple poisoned pairs) fall outside this single-pair model and are deferred to Section VI.

Assumption A4 concerns expected continuation values, not individual fading realizations. Fast, action-independent fading can have high sample variance while largely averaging out in $U(a)$; C remains small when competing resource-allocation actions lead to similar next-state value distributions. Persistent action-coupled congestion, queue buildup, mobility, or handover can instead enlarge $U(a) - U(a^*)$ and require a larger C . An operator can estimate C conservatively from clean rollouts by taking an upper quantile of positive action-conditioned continuation differences normalized by Δ_c^{max} . A nonpositive L_α makes Proposition 1 vacuous. The detector can still be evaluated empirically, but the proposition supplies no certificate for that deployment.

Proposition 1 gives a tabular bound; the trained-network counterpart $\rho_{\text{NN}}(\theta) := \Delta(s_{\text{trig}}; \theta)/\Delta_{\text{baseline}}(\theta)$ and the dispersion ratio $\eta(\theta) := \rho_{\text{NN}}(\theta)/\rho_{\text{tab}}^b$ bridge the bound to NN ratios, where $\rho_{\text{tab}}^b := (r_p - r_c^{\text{sup}})/\bar{\Delta}_c \approx 75$ is the unattenuated tabular reference at $\alpha=1$, $C=0$ evaluated at the typical clean Q-gap $\bar{\Delta}_c \approx 0.12$ (median) from Sec. IV-F (using the looser empirical upper bound $\Delta_c^{\text{max}} \lesssim 0.36$ from the Remark would give $\rho_{\text{tab}}^b=25$, which is a conservative reference). The 9 standard DQN slicing trojans (Table 3) exhibit $\rho_{\text{NN}} \in [12.4, 67.8]$ ($\eta \in [0.17, 0.91]$); the maximum

clean envelope $\rho_{\text{clean}} \leq 4.13$ on the 21 calibration models gives a separation factor $12.4/4.13 \approx 3.0\times$, so any threshold in $(4.13, 12.4)$ separates clean and trojan, and the calibrated $\tau_{99.9\%} = 4.66$ from Sec. IV-F lies inside this interval. The detection guarantee thus rests on the empirical separation rather than on a deterministic upper bound on ρ_{NN} . Per-family ranges for DDQN, PPO/A2C, and SAC under the alternate normalizers σ_π and $Z_F=1$ are reported in Table 13.

In our stealth variant, the regularizer is applied on clean inputs and does not explicitly suppress the trigger-state Q-gap; empirically, the trigger gap remains large at $\rho = 37.7\times$, well above the detection threshold (Table 3).

Let $\rho_c = \max_i \rho_i$ denote the maximum gap ratio observed across clean models, and $\rho_t = \min_j \rho_j$ denote the minimum gap ratio across trojan models. GRADFA separates clean from trojan models when

$$\rho_t > \tau_\alpha > \rho_c, \quad (6)$$

where τ_α is the adaptive threshold from (18). In our experiments, $\rho_c = 4.13$, $\tau_\alpha = 4.66$, and $\rho_t = 12.37$ for the DQN slicing profile, yielding a separation margin of $\rho_t/\tau_\alpha = 2.65\times$.

B. Extension to PPO Models

The gap ratio principle above assumes access to Q-values, which limits it to value-based methods like DQN and DDQN. For PPO models with continuous action spaces, Q-values are unavailable. We adapt the detection principle by replacing the Q-gap with the action shift magnitude. For each feature group g , we measure the L2 distance between the model's deterministic action output on swept observations and its baseline output:

$$\delta_g = \max_{v \in \mathcal{V}} \frac{1}{|\mathcal{D}|} \sum_{\mathbf{s} \in \mathcal{D}} \|\pi_\theta(\mathbf{s}|_{g \leftarrow v}) - \pi_\theta(\mathbf{s})\|_2, \quad (7)$$

where $\pi_\theta(\mathbf{s})$ denotes the deterministic action output. The action shift ratio is then $\rho_g^{\text{PPO}} = \delta_g/\sigma_\pi$, where $\sigma_\pi = \sqrt{\text{tr}(\text{Cov}[\pi_\theta(\mathbf{s})])}$ is the total standard deviation of the action vector computed on un-swept baseline observations. The default threshold for PPO is lower than the DQN default ($\tau_1 = 3.0$) because behavioral cloning of the policy head produces a weaker backdoor signal than replay-buffer poisoning of the Q-network. The same statistic applies to A2C and other on-policy actor-critic methods that expose a Gaussian (or softmax) policy head.

C. Extension to Off-Policy Actor-Critic: Critic Q-Gap Probe

Off-policy actor-critic algorithms such as SAC require a different statistic. Their tanh-squashed Gaussian actor with entropy regularization is responsive to observation perturbations, so the action-shift statistic from (7) produces clean ratios that overlap the trojan range. SAC's twin Q-network architecture, however, exposes a different attack surface: a replay-buffer poisoning attacker injects

$(\mathbf{s}_{\text{trig}}, \mathbf{a}^*, r_p, \mathbf{s}')$ transitions, causing the critic to converge toward $Q_1(\mathbf{s}_{\text{trig}}, \mathbf{a}^*) \gtrsim r_p$ at the trigger. The critic-side spike is analogous to DQN's Q-gap, but defined over a continuous action space.

Since the action space is continuous, we discretize the actor's $[0, 1]^3$ output cube into K representative query vectors $\mathcal{A}_K = \{\mathbf{a}_1, \dots, \mathbf{a}_K\}$ that span one-hot, two-hot, corners, centroid, and axis-aligned 80/10/10 concentrations. The default $K=12$ covers all three trojan payload geometries defined in Section III-C. The per-state Q-gap is $\Delta^{\text{SAC}}(\mathbf{s}) = \max_k Q_1(\mathbf{s}, \mathbf{a}_k) - Q_1^{(2)}(\mathbf{s})$.

For DQN we normalized the swept gap by the baseline gap to produce a model-invariant ratio. For SAC this normalization fails because critic-poisoning fine-tuning inflates Q-magnitudes globally, lifting the baseline denominator and cancelling the trojan signal in the ratio (the absolute trigger spike stays large while the baseline inflates with poison reward). We therefore report the absolute statistic

$$\Phi^{\text{SAC}} = \max_{g \in \mathcal{G}} \max_{v \in \mathcal{V}} \frac{1}{|\mathcal{D}|} \sum_{\mathbf{s} \in \mathcal{D}} \Delta^{\text{SAC}}(\mathbf{s}|_{g \leftarrow v}). \quad (8)$$

Detection flags suspect models whose Φ^{SAC} exceeds a fixed-scalar threshold set above the maximum clean value observed across the calibration set (in our evaluation, $\tau_{\text{SAC}} = 2.0$, placed in the strict gap between max clean 1.54 and min trojan 3.87 at $K = 12$; Table 13). Any choice within this gap yields identical 100% TPR / 0% FPR.

D. Proposed Algorithm

With the three family-specific statistics defined (Q-gap ratio for DQN/DDQN, action shift for PPO/A2C, and absolute critic Q-gap for SAC), we now describe the unified detection procedure that wraps them. GRADFA exploits the O-RAN observation structure by sweeping semantic feature groups rather than individual features. The 15-dimensional observation is partitioned into five groups corresponding to O-RAN KPM categories, one feature index per slice: traffic demand ($\{0, 5, 10\}$), channel quality ($\{1, 6, 11\}$), PRB allocation ($\{2, 7, 12\}$), throughput ratio ($\{3, 8, 13\}$), and SLA satisfaction ($\{4, 9, 14\}$). The sweep grid is $\mathcal{V} = \{0.0, 0.1, \dots, 1.0\}$ (11 points). The two-tier sweep structure (single-group then pairwise) is shared across algorithm families; only the family-specific pair $(\text{score}(g, v), Z)$ defined below changes.

The trusted mapping is obtained from the onboarding descriptor and verified by the conformance adapter described in Section III-A. Automatically generated groups preserve the declared within-slice field order across repeated slice blocks. Partial mappings sweep only known coordinates, while a mapping-free fallback treats each coordinate as a singleton; neither fallback is assumed to retain the accuracy of a correct semantic mapping. A descriptor that fails conformance therefore produces an “unverifiable/quarantine” outcome, not “clean.”

To defend against distributed triggers that spread across multiple feature groups, GRADFA employs two-tier detection:

- **Tier 1 (single-group sweep):** Sweep each of $|\mathcal{G}| = 5$ groups independently. Detects standard triggers. Threshold τ_1 .
- **Tier 2 (pairwise-group sweep):** Sweep all $\binom{|\mathcal{G}|}{2} = 10$ pairs simultaneously. Detects distributed triggers. Threshold $\tau_2 = 3\tau_1$.

Algorithm 1 states the procedure with symbolic thresholds τ_1, τ_2 . For exposition we use a fixed default ($\tau_1 = 3, \tau_2 = 9$) inherited from the gap-ratio scaling analysis in Section IV-A. For DQN/DDQN evaluations the thresholds are instead set by the t -distribution calibration of Section IV-F (Tier 1: τ_α from (18); Tier 2: $3\tau_\alpha$); for A2C, SAC, and PPO evaluations we use family-specific fixed-scalar thresholds placed inside the strict clean-trojan separation gap.

For a DQN/DDQN model that clears both Q-gap tiers, we apply a conservative cross-signal guard using quantities already computed for attribution. Let $\mathbf{d}_a = N^{-1} \sum_{\mathbf{s} \in \mathcal{D}} \nabla_{\mathbf{s}} Q_\theta(\mathbf{s}, a)$ and let z_a be the modified-median-absolute-deviation score of $1 - \cos(\mathbf{d}_a, (|\mathcal{A}| - 1)^{-1} \sum_{b \neq a} \mathbf{d}_b)$ across actions. The guard flags only if the Tier-1 sweep-localized action $\hat{a}^*$ equals $\arg \max_a z_a$ and $\max_a z_a > 3$. It is an independent confirmation channel rather than a third sweep tier. Section V-F evaluates the fixed rule over 60 clean DQN models, three tasks, and ten observation banks; no trojan score selects the rule or threshold.

The algorithm is parameterized by a per-family pair $(\text{score}(g, v), Z)$: $\text{score}(g, v)$ is the mean model response when feature group g is overwritten with sweep value v , and Z is a family-specific normalizer that makes the ratio $\rho_g = \max_v \text{score}(g, v)/Z$ comparable across models. We write $\mathbf{s}|_{g \leftarrow v}$ for the observation $\mathbf{s}$ with features in g set to v , and overload g to mean either a single group or a pair $\{g_i, g_j\}$. The three families instantiate (score, Z) as follows.

For DQN and DDQN, the score and normalizer instantiate the Q-gap (Sec. IV-A, eq. (4)):

$$\begin{aligned} \text{score}(g, v) &= \frac{1}{N} \sum_{\mathbf{s} \in \mathcal{D}} [\max_a Q_\theta(\mathbf{s}|_{g \leftarrow v}, a) - Q_\theta^{(2)}(\mathbf{s}|_{g \leftarrow v})], \quad (9) \\ Z &= \frac{1}{N} \sum_{\mathbf{s} \in \mathcal{D}} [\max_a Q_\theta(\mathbf{s}, a) - Q_\theta^{(2)}(\mathbf{s})] = \Delta_{\text{baseline}}. \end{aligned}$$

For PPO and A2C, the score and normalizer instantiate the action-shift statistic (Sec. IV-B, eq. (7)):

$$\begin{aligned} \text{score}(g, v) &= \frac{1}{N} \sum_{\mathbf{s} \in \mathcal{D}} \|\pi_\theta(\mathbf{s}|_{g \leftarrow v}) - \pi_\theta(\mathbf{s})\|_2, \quad (10) \\ Z &= \sigma_\pi = \sqrt{\text{tr}(\text{Cov}_{\mathbf{s} \in \mathcal{D}}[\pi_\theta(\mathbf{s})])}. \end{aligned}$$

For SAC, the score and normalizer instantiate the absolute critic Q-gap (Sec. IV-C, eq. (8)):

$$\begin{aligned} \text{score}(g, v) &= \frac{1}{N} \sum_{\mathbf{s} \in \mathcal{D}} [\max_k Q_1(\mathbf{s}|_{g \leftarrow v}, \mathbf{a}_k) - Q_1^{(2)}(\mathbf{s}|_{g \leftarrow v})], \quad (11) \\ Z &= 1 \quad (\text{absolute test; rationale in Sec. IV-C}). \end{aligned}$$

Lemma 1 (Family-aware separation criterion). *Fix $F \in \{\text{DQN/DDQN}, \text{PPO/A2C}, \text{SAC}\}$ with the family-specific responsiveness functional $\Psi_F(\theta, \mathbf{s}, g, v) \geq 0$ and normalizer $Z_F(\theta) \in (0, \infty)$ instantiated in Sec. IV-D (SAC takes $Z_F=1$, so ρ_F is an unnormalized detection score). Define the unified detection score*

$$\rho_F(\theta) := \max_{g \in \mathcal{G}, v \in \mathcal{V}} \frac{\frac{1}{N} \sum_{\mathbf{s} \in \mathcal{D}} \Psi_F(\theta, \mathbf{s}, g, v)}{Z_F(\theta)}, \quad (12)$$

and let $\mathcal{C}_F^{\text{cal}}$ and $\mathcal{T}_F^{\text{cal}}$ denote the nonempty finite clean and trojan model classes covered by the calibration set, with empirical clean envelope

$$\rho_F^{\text{clean}, \max} := \max_{\theta_c \in \mathcal{C}_F^{\text{cal}}} \rho_F(\theta_c) < \infty. \quad (13)$$

Assume that every trojan in $\mathcal{T}_F^{\text{cal}}$ satisfies a trigger-response floor:

- (B1) *Trigger amplification. For each $\theta_t \in \mathcal{T}_F^{\text{cal}}$ there exists $(g^*, v^*) \in \mathcal{G} \times \mathcal{V}$ with*

$$\frac{1}{N} \sum_{\mathbf{s} \in \mathcal{D}} \Psi_F(\theta_t, \mathbf{s}, g^*, v^*) \geq \kappa_F(\theta_t) M_F(\theta_t), \quad (14)$$

where $M_F(\theta_t) > 0$ is the mechanism-specific nominal attack magnitude (see the Remark below) and $\kappa_F(\theta_t) > 0$ is a model-specific constant absorbing family-dependent attenuation.

Define the uniform trojan floor

$$L_F := \min_{\theta_t \in \mathcal{T}_F^{\text{cal}}} \frac{\kappa_F(\theta_t) M_F(\theta_t)}{Z_F(\theta_t)}. \quad (15)$$

- (B2) *Strict separation. $L_F > \rho_F^{\text{clean}, \max}$.*

Then every fixed threshold $\tau_F \in (\rho_F^{\text{clean}, \max}, L_F)$ strictly separates the calibration clean and trojan classes:

$$\rho_F(\theta_c) < \tau_F < \rho_F(\theta_t), \quad \forall \theta_c \in \mathcal{C}_F^{\text{cal}}, \quad \forall \theta_t \in \mathcal{T}_F^{\text{cal}}. \quad (16)$$

The proof is deferred to Appendix B. Beyond the inequality, the lemma gives a single criterion that makes the three heterogeneous statistics (ratio-normalized, σ_π -normalized, unnormalized) comparable, so the per-family thresholds belong to one framework rather than three ad hoc choices.

The lemma does not establish B1 or B2 for unseen model populations. In particular, B2 assumes the strict separation that the lemma then propagates to any threshold inside the interval; it is a calibration-set criterion, not a proof of out-of-sample separability. We therefore validate thresholds on held-out clean models in Section V-D and report failure cases explicitly.

The trigger response floor (B1) is a sufficient condition for the lemma to apply, not a consequence of the attack design. We treat it as an empirical hypothesis that must be checked per family on the calibration set, and instantiate M_F and κ_F differently per family to identify the dominant mechanism in each case:

- *Value-based (DQN/DDQN)*. We take $M_F = r_p - r_c^{\text{sup}}$ as the nominal poisoned-reward margin. Under Proposition 1, this margin yields a tabular trigger Q-gap floor when the conditional poisoning rate at (s_{trig}, a^*) is high and the continuation gap is controlled. The constant $\kappa_F(\theta_t)$ then absorbs the conditional-poisoning attenuation $\alpha \leq 1$ from the Remark on the threat model, the continuation correction $C\gamma$, the normalizer conversion $\Delta_{\text{baseline}}/\Delta_c^{\text{max}}$, the function-approximation dispersion ratio $\eta(\theta_t)$ defined after Proposition 1, and the fraction of $\mathcal{D}$ for which the sweep (g^*, v^*) falls in the trigger neighborhood.
- *Off-policy actor-critic (SAC)*. $M_F = r_p - r_c^{\text{sup}}$ from critic poisoning. (B1) holds when the poisoned updates affect the critic used to compute Ψ_F ; $\kappa_F(\theta_t)$ absorbs attenuation from the twin-Q minimum in the SAC target, the entropy term, target-network lag, and which of the twin critics receives the poisoned reward signal.
- *On-policy actor-gradient (PPO/A2C)*. $M_F = \|\mathbf{a}^* - \pi_c(s_{\text{trig}})\|_2$ from behavioral cloning. (B1) holds when the behavioral cloning (BC) component dominates the clean-policy retention residual; $\kappa_F(\theta_t)$ summarizes the retained fraction of this nominal action shift after PPO clipping, entropy regularization, advantage gradients, and function approximation.

We introduce the observed calibration-set trojan floor L_F^{cal} , defined as

$$L_F^{\text{cal}} := \min_{\theta_t \in \mathcal{T}_F^{\text{cal}}} \rho_F(\theta_t) \quad (17)$$

By (B1) it cannot fall below the analytical floor: $L_F^{\text{cal}} \geq L_F$, so the analytical L_F is a possibly conservative lower estimate of what is actually observed. The per-family L_F^{cal} values reported in Table 13 are 12.4, 13.5, 1.7, 3.9, 2.5 for DQN, DDQN, A2C, SAC, PPO respectively, and exceed the corresponding observed clean envelopes ($\rho_{\text{DQN}}^{\text{clean,max}} = 4.13$ on the slicing profile, $\rho_{\text{DDQN}}^{\text{clean,max}} = 3.76$, $\rho_{\text{A2C}}^{\text{clean,max}} = 1.35$, $\rho_{\text{SAC}}^{\text{clean,max}} = 1.54$ at $K=12$, $\rho_{\text{PPO}}^{\text{clean,max}} = 2.14$ on the reward-gated set; details in Section V-J). The empirical observation $\rho_F^{\text{clean,max}} < L_F^{\text{cal}}$ in every family directly verifies the separation conclusion (16) on the calibration set, independently of how tight the analytical constants in (B1) are. Fig. 1 illustrates the shared detection flow.

Complexity. With $N=|\mathcal{D}|$ rollout observations, $G=|\mathcal{G}|$ feature groups, $V=|\mathcal{V}|$ sweep values, and F the cost of one forward pass, the two-tier sweep (GV single-group and $\binom{G}{2}V$ pairwise configurations, each a batched pass over $\mathcal{D}$) runs in $O(G^2 V N F)$ time and $O(Nd + |\theta|)$ space, where d is the observation dimension and $|\theta|$ the parameter count; no gradients are stored, and SAC adds a factor K for its candidate-action set. The conditional DQN/DDQN guard adds $O(|\mathcal{A}| N F)$ gradient work and a batched $O(Nd)$ buffer only for models below both Q-gap thresholds; on the Section V-A workstation it adds 0.038 s to the 0.110 s

Algorithm 1 GRADFA two-tier feature-sweep detector with a conditional value-based guard. Family responses are Q-gap for DQN/DDQN, action shift for PPO/A2C, and absolute critic Q-gap ($Z=1$) for SAC.

Require: Suspect model θ , environment $\mathcal{E}$, feature groups $\mathcal{G}$, sweep grid $\mathcal{V}$, thresholds τ_1, τ_2 , family pair (score, Z)

Ensure: Verdict $\in \{\text{CLEAN}, \text{TROJAN}\}$, with $(\hat{g}^*, \hat{v}^*, \hat{a}^*, \hat{\rho})$ if Trojan

```

1: // Stage 0: Baseline rollout (shared by both tiers)
2:  $\mathcal{D} \leftarrow$  rollout  $N=1000$  obs from  $\mathcal{E}$  under the suspect policy  $\theta$ 
3: compute the family normalizer  $Z$  over  $\mathcal{D}$ 
4: // Tier 1: Single-group sweep (standard triggers)
5: for each group  $g \in \mathcal{G}$  do
6:    $v_g^* \leftarrow \arg \max_{v \in \mathcal{V}} \text{score}(g, v)$  (best sweep value for  $g$ )
7:    $\rho_g \leftarrow \text{score}(g, v_g^*) / Z$  (gap ratio for  $g$ )
8: end for
9:  $g^\dagger \leftarrow \arg \max_{g \in \mathcal{G}} \rho_g$ 
10: if  $\rho_{g^\dagger} > \tau_1$  then
11:    $\hat{g}^* \leftarrow g^\dagger$ ;  $\hat{v}^* \leftarrow v_{g^\dagger}^*$ ;  $\hat{\rho} \leftarrow \rho_{g^\dagger}$ 
12:    $\hat{a}^* \leftarrow \text{TARGETACTION}(\theta, \hat{g}^*, \hat{v}^*)$ 
13:   return TROJAN(tier = 1,  $\hat{g}^*, \hat{v}^*, \hat{a}^*, \hat{\rho}$ )
14: end if
15: // Tier 2: Pairwise-group sweep (distributed triggers)
16: for each pair  $g_{ij} = \{g_i, g_j\}$ ,  $i < j$  do
17:    $v_{ij}^* \leftarrow \arg \max_{v \in \mathcal{V}} \text{score}(g_{ij}, v)$  (single  $v$  applied to both groups in  $g_{ij}$ )
18:    $\rho_{ij} \leftarrow \text{score}(g_{ij}, v_{ij}^*) / Z$ 
19: end for
20:  $(i^\dagger, j^\dagger) \leftarrow \arg \max_{(i,j)} \rho_{ij}$ 
21: if  $\rho_{i^\dagger j^\dagger} > \tau_2$  then
22:    $\hat{g}^* \leftarrow \{g_{i^\dagger}, g_{j^\dagger}\}$ ;  $\hat{v}^* \leftarrow v_{i^\dagger j^\dagger}^*$ ;  $\hat{\rho} \leftarrow \rho_{i^\dagger j^\dagger}$ 
23:    $\hat{a}^* \leftarrow \text{TARGETACTION}(\theta, \hat{g}^*, \hat{v}^*)$ 
24:   return TROJAN(tier = 2,  $\hat{g}^*, \hat{v}^*, \hat{a}^*, \hat{\rho}$ )
25: end if
26: // Conditional DQN/DDQN cross-signal guard
27: if value-based model and both tiers are below threshold then
28:    $a_\nabla \leftarrow \arg \max_a z_a$  from signed input-gradient directions  $\mathbf{d}_a$ 
29:   if  $a_\nabla = \text{TARGETACTION}(\theta, g^\dagger, v_{g^\dagger}^*)$  and  $\max_a z_a > 3$  then
30:     return TROJAN(channel = cross-signal guard, action =  $a_\nabla$ )
31:   end if
32: end if
33: end if
34: return CLEAN
35: Helper TARGETACTION( $\theta, g, v$ ): return the action most amplified by the trigger pattern on  $\mathcal{D}$ . For DQN/DDQN,  $\arg \max_a \frac{1}{N} \sum_{s \in \mathcal{D}} Q_\theta(s|g \leftarrow v, a)$ ; for PPO/A2C, the largest-shift coordinate of  $\frac{1}{N} \sum_s [\pi_\theta(s|g \leftarrow v) - \pi_\theta(s)]$ ; for SAC,  $\arg \max_k \frac{1}{N} \sum_s Q_1(s|g \leftarrow v, \mathbf{a}_k)$ .
```

sweep (10-run means), excluding the shared rollout. Using only forward passes, GRADFA avoids the per-target trigger reconstruction of Neural Cleanse and BIRD ($O(T|\mathcal{A}|F)$ over $|\mathcal{A}|$ candidate actions), the source of their order-of-magnitude-higher runtimes (Table 6).

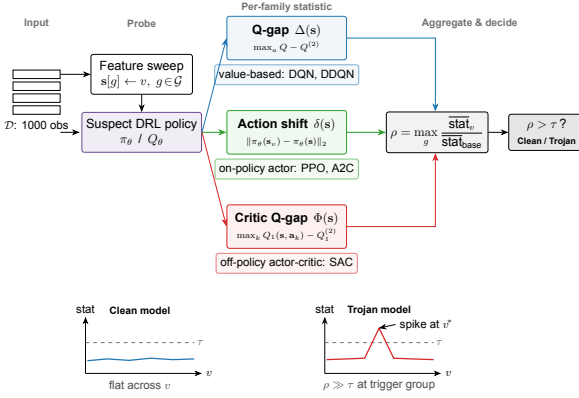


FIGURE 1. GRADFA detection method. The observation vector is swept per semantic feature group and passed through the suspect model. The per-state statistic is selected by paradigm: value-based agents (DQN, DDQN) use the Q-value gap Δ (Section IV-A; pseudocode in Algorithm 1), on-policy actor agents (PPO, A2C) use the deterministic-action shift magnitude δ (Section IV-B), and off-policy actor-critic agents (SAC) use the absolute critic Q-gap Φ over K candidate actions (Section IV-C). The DQN/DDQN and PPO/A2C branches normalize by a baseline to form a gap ratio, whereas the SAC branch reports the absolute max gap (Section IV-C); for compact notation the diagram labels the test score as ρ across branches, with $\rho > \tau$ flagging a suspect. Clean models produce flat profiles; trojan models exhibit a spike at the trigger value v^* .

The sweep contains $V[G + G(G - 1)/2]$ configurations: 165 at $(G, V) = (5, 11)$, 1,155 at $(10, 21)$, 4,410 at $(20, 21)$, and 17,220 at $(40, 21)$. Relative to the measured 0.12 s sweep time at 165 configurations, these counts imply first-order central-processing-unit (CPU) projections of 0.84, 3.21, and 12.52 s, respectively, before accounting for a different network forward cost. Configurations are processed sequentially, so they do not accumulate in memory. In float32, the persistent probe storage is approximately $4Nd + 4|\theta|$ bytes (plus one batched activation/output buffer); increasing the observation dimension from 15 to 256 at $N = 1000$ raises observation-bank storage from 0.06 to 1.02 megabytes (MB). For SAC, compute grows linearly with candidate count K ; Table 12 reports the measured $K \in \{6, 12, 24\}$ ablation. These projections characterize complexity only and do not establish detection accuracy for untested architectures.

For an LSTM- or GRU-based policy, a temporal extension warms up and preserves the hidden state, then sweeps a feature group over a contiguous subwindow of an H -step trajectory. Tier 1 searches group, start, duration, and value; Tier 2 searches pairs of group/window patterns using a maximum or time-integrated response. The cost multiplies by the number of temporal windows. For multi-agent policies, groups are indexed by agent and the probe targets the joint policy or centralized critic while other histories remain fixed. Section V-G tests a three-step RecurrentPPO case and a centralized two-cell DQN. Graph, transformer, decentralized, independently learning, and larger multi-agent policies remain untested.

E. Properties

The compute is bounded: Tier 1 evaluates $|\mathcal{G}| \times |\mathcal{V}| = 5 \times 11 = 55$ sweep configurations and Tier 2 adds $\binom{|\mathcal{G}|}{2} \times |\mathcal{V}| = 110$ pairwise configurations, each a single batched forward pass over the $|\mathcal{D}|=1000$ -observation set, totalling 1.65×10^5 forward evaluations (approximately 7.6 billion floating-point operations (GFLOP) on the 2×128 network) with no backpropagation and ~ 0.48 s per model on a single Apple M4 CPU core (Table 6, mean of 6 runs; breakdown in Section V-L). A value-based model below both tiers incurs the additional 0.038 s cross-signal check, for approximately 0.52 s including the shared rollout. Unlike Behavioral Comparison (D1), GRADFA analyzes a single suspect model, which matters for marketplace deployment where the operator may not have a trusted clean reference. The verdict is interpretable per feature group: GRADFA returns the suspect group $\hat{g}^*$, the suspect target action $\hat{a}^*$, and the gap ratio ρ , which feed directly into severity assessment and remediation (Sections IV-G2–IV-G3).

F. Statistical Calibration

Algorithm 1 uses fixed thresholds τ_1 and τ_2 , which may not generalize across different xApp tasks. To make detection task-adaptive, we calibrate the detection threshold against a distribution of legitimate models and report a p -value for each suspect model.

Given n clean models ($n = 21$), we compute $\rho_i = \max_g \rho_g$ for each model i , yielding a calibration distribution $\{\rho_1, \dots, \rho_n\}$ with sample mean $\hat{\mu}$ and sample standard deviation $\hat{\sigma}$. For a target confidence level α (e.g., 99.9%), the adaptive threshold is

$$\tau_\alpha = \hat{\mu} + t_{n-1}^{-1}(\alpha) \cdot \hat{\sigma}, \quad (18)$$

where $t_{n-1}^{-1}(\alpha)$ denotes the inverse cumulative distribution function (CDF) of the Student's t -distribution with $n-1$ degrees of freedom.

For a suspect model with gap ratio ρ_s , the standardized test statistic and one-sided p -value are

$$t = \frac{\rho_s - \hat{\mu}}{\hat{\sigma}}, \quad p = 1 - F_{t,df}(t), \quad (19)$$

where $F_{t,df}$ denotes the CDF of Student's t -distribution with $df = n - 1$. The model is flagged if $\rho_s > \tau_\alpha$, as computed by (19). In our evaluation, the calibration via (18) yields $\hat{\mu} = 2.32$, $\hat{\sigma} = 0.66$, $\tau_{99.9\%} = 4.66$ for the DQN slicing profile. The raw per-state Q-gap Δ_{baseline} on these 21 models spans $[0.06, 0.36]$ (mean 0.14, median 0.12); the ratio normalizes this scale away.

We apply t -calibration to the Q-gap ratio statistic for DQN (where the 21 clean slicing models pass a Kolmogorov–Smirnov normality test with $p = 0.63$) and to DDQN (clean range $[1.20, 3.76] \times$ at $n_c=21$ supporting the same parametric assumption). For A2C action-shift, SAC absolute critic-gap, and PPO action-shift, the clean and trojan ranges are strictly disjoint, so we report a fixed scalar threshold placed inside the gap rather than fitting a parametric distribution; the scalar

choice within the strict separation does not affect the binary detection outcome.

Because those calibration-set results can be optimistic, Section V-D adds a separate, nonparametric validation. Within each family and task, approximately 75% of clean models form a calibration partition; the threshold is the maximum calibration score, without using any trojan score, and the remaining clean models estimate held-out FPR. All trojan models are used only as test samples. We repeat the stratified split 1,000 times to report threshold and FPR stability. The maximum-score rule is deliberately distribution-free but has finite rank resolution $1/(n_{\text{cal}} + 1)$, so its confidence intervals remain wide for the smaller families.

G. Proposed Verification Pipeline for O-RAN xApp Lifecycle

We embed GRADFA in a three-stage pipeline (detect, assess, remediate; Fig. 2) that adds severity assessment and remediation on top of the binary clean-or-compromised verdict.

1) Stage 1-Statistical Detection

When a vendor submits an xApp, GRADFA runs automatically against the family-appropriate threshold (a t -calibrated τ_α for DQN/DDQN, a fixed-scalar threshold for A2C/SAC/PPO; Section IV-F). The output includes a p -value (when t -calibrated), suspect feature group $\hat{g}^*$, and target action $\hat{a}^*$. When GRADFA flags a suspect, its output ($\hat{g}^*, v^*$) doubles as a synthetic trigger that re-weaponizes D4 Spectral Signatures: the latter fails as a standalone detector (Table 6) because it lacks knowledge of the trigger pattern, but as a confirmation signal, the method adds defense-in-depth at no extra probing cost. We report this Spectral confirmation as a deployment hook rather than a quantitative result; a full TPR/FPR characterization across the 118 trojan models (and 9 DQN stealth variants) is left to future work.

2) Stage 2-Network-Impact-Aware Severity Assessment

Rather than issuing a binary reject/accept decision, Stage 2 quantifies the operational risk of the detected compromise using 3GPP TS 28.554-aligned KPIs per slice: mean throughput, SLA violation rate, and a fairness index (we report the same coefficient-of-variation-based form used in the training reward); per-slice tail metrics (e.g., P5 throughput, P95 latency) are exposed by the simulator and can be enabled when the operator's SLA defines tail-percentile clauses. We classify each slice as a victim (throughput drop $> 5\%$) or winner (throughput gain from misallocated resources). The Stage 2 evaluation in Section V-H reports the three core KPIs and the victim/winner labels directly; mapping them into a single composite severity score (e.g., 0–100 with LOW/MEDIUM/HIGH/CRITICAL bands) is operator-policy specific and is left as a deployment-time choice.

3) Stage 3-Targeted Remediation

Stage 3 offers two remediation strategies.

Strategy A (runtime action masking) blocks the suspect action whenever observation features in $\hat{g}^*$ approach v^* , by setting $Q(s, \hat{a}^*) = -\infty$ to force the second-best action. This is instant and requires no retraining.

Strategy B (output-layer fine-tuning (FT)) freezes the hidden layers and injects suppression transitions ($s_{\text{trig}}, \hat{a}^*, r_{\text{pen}}, s'$) with $r_{\text{pen}} = -5.0$ at a 15% rate over 10K fine-tuning steps, reducing ASR from 100% to 0%. The held-out KPI evaluation in Section V-I shows that this suppresses the target action but does not fully restore trigger-conditioned policy utility. These settings invert the attack: the negative r_{pen} (opposite the attacker's $r_p=10$) drives the trojan action's trigger Q-value back below its competitors, the 15% rate (near the poison ratio) makes the correction dominate while clean transitions limit utility drift, and 10K steps suffice since only the output layer is updated.

V. Simulation Results

We evaluate the three-stage pipeline of Fig. 2 end-to-end, and organize this section around its stages. For Stage 1 (detection), we first establish core accuracy on the DQN slicing task including the behavior-preserving stealth variant (Section V-B), evaluate feature-mapping robustness and independent held-out calibration (Sections V-C and V-D), then benchmark against five adapted supervised-domain defenses and BIRD (Section V-E), probe robustness to adaptive attackers (Section V-F), generalize the detector across five DRL families and three xApp tasks (Section V-J), and test transfer across six observation environments (Section V-M). For Stage 2 (severity), we quantify the network-level harm a trojan inflicts if left undetected (Section V-H). For Stage 3 (remediation), we measure target-action suppression, residual policy utility, and repeated-trigger behavior over 2,000 steps (Section V-I). Finally, we report the per-model runtime overhead (Section V-L) and a complementary post-deployment multi-xApp defense-in-depth layer (Section V-K).

A. Simulation Settings

The environment is a custom O-RAN network-slicing setup based on CoO-RAN [4]: 3 slices, $N_{\text{PRB}} = 100$ PRBs, 15-dimensional observations, 36 discrete actions, and 200-step episodes. The per-PRB bandwidth $B_{\text{PRB}} = 360$ kHz (12 subcarriers $\times$ 30 kHz subcarrier spacing) follows the 3GPP New Radio (NR) numerology [34]. The remaining environment parameters are set by us for this study rather than fitted to measured data. Traffic demand per slice is a clipped Gaussian $d_i \sim \mathcal{N}(\mu_i, \sigma_i^2)$ on $[0, 1]$, with means and spreads we choose to give the three slices distinct load profiles (eMBB: $\mu=0.6, \sigma=0.2$; URLLC: $\mu=0.3, \sigma=0.1$; mMTC: $\mu=0.4, \sigma=0.15$). Channel quality (SINR) is drawn per step from a Gaussian with mean 15 dB and standard deviation 5 dB, clipped to $[0, 30]$ dB and normalized to $[0, 1]$,

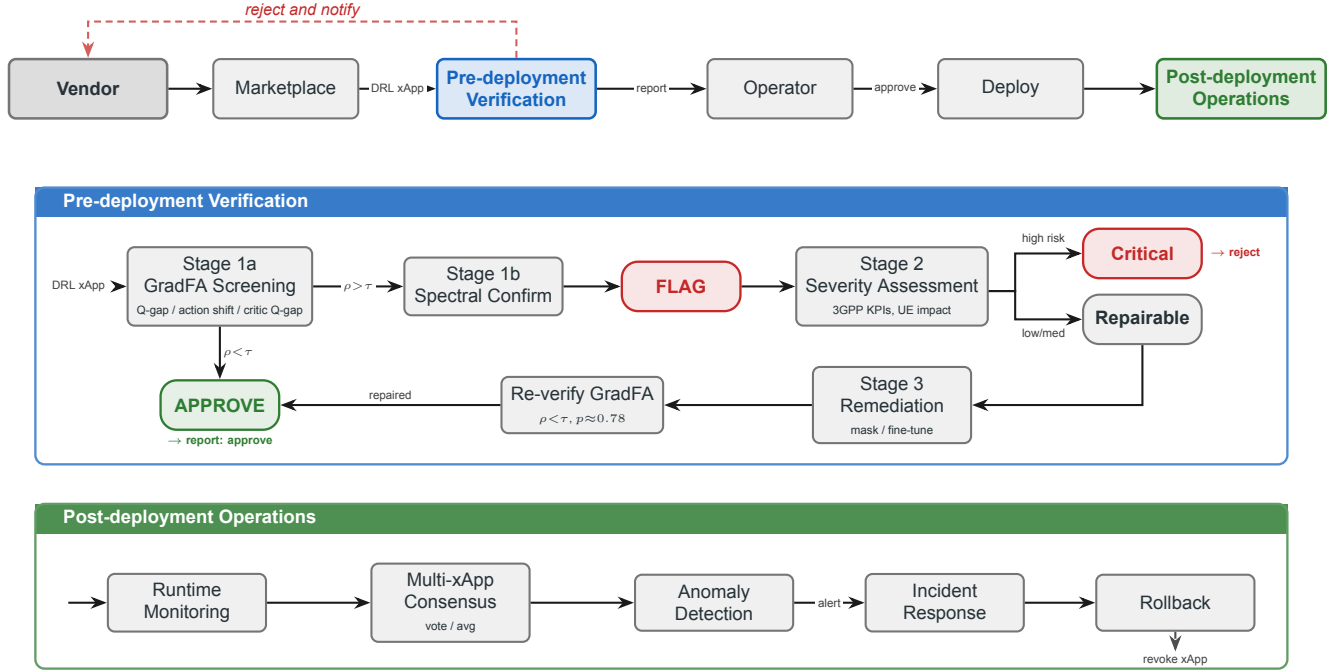


FIGURE 2. GRADFA verification pipeline integrated into the O-RAN xApp lifecycle. A vendor-submitted DRL xApp enters Stage 1a (GRADFA Screening, family-appropriate statistic from Fig. 1); $p > \tau$ flags the model, with an optional Stage 1b Spectral confirmation (not evaluated in this work; Section IV-G1). Stage 2 reports per-slice 3GPP KPIs with CRITICAL/REPAIRABLE bands as one example operator routing policy (Section IV-G2). Stage 3 applies action masking or output-layer fine-tuning, after which GRADFA is re-run on the repaired model (indicative $p \approx 0.90$ corresponds to the regenerated DQN slicing/high_load repair in Table 10

). Post-deployment hooks (multi-xApp consensus, runtime anomaly detection, incident response, rollback) are listed for completeness; only multi-xApp consensus is evaluated here (Section V-K).

a lightweight stand-in for the fast-fading and shadowing of the 3GPP channel model [35]; the mean and spread are set by us to represent a typical operating point rather than drawn from it. We likewise set the remaining constants of the environment model (Section III): PRB-allocation grid step 0.1 with floor $f_{\min} = 0.05$; demand utilization $u_i = \min(1, 1.5 d_i)$; throughput ratio $\eta_i = \min(T_i/T_i^{\text{SLA}}, 2)/2$; latency constants $c_L = 0.8$, $d_0 = 0.5$, $L_i^{\text{base}} = 0.3 L_i^{\text{SLA}}$; and reward weights $w_{\text{URLLC}} = 2$, $w_{\text{EMBB}} = 1$, $w_{\text{mMTC}} = 0.5$ (ordering per ITU-R [32]) with violation penalty $\beta = 0.5$ and fairness weight $\lambda = 0.3$. Each episode generates a fresh 200-step trace with independent traffic and channel realizations, ensuring that no two episodes share the same network conditions; no external datasets are used.

Each model is evaluated over 50 episodes (10,000 steps total) with fixed random seeds for reproducibility. Attack success rate (ASR) is computed as the fraction of triggered steps where the agent selects the target malicious action. Clean performance is measured as mean episodic reward averaged over the 50 episodes. For GRADFA detection, 1,000 observations are sampled from the environment under the suspect xApp's own policy (5 episodes of 200 steps each) and used to compute the baseline Q-gap and perform feature sweeps (11 sweep points per group, from 0.0 to 1.0).

The Q-gap ratio is approximately invariant to the observations used to compute the baseline, because it is a property of

the trained policy rather than the deployment environment. Section V-M empirically validates both attack transfer and GRADFA detection across six heterogeneous environments, including three 3GPP-calibrated channel models, the Colosseum testbed trace, and a real 5G network trace.

The evaluation covers 280 models across five DRL families, summarized in Table 2. The main detection table (Table 13) reports 262 of these (144 clean + 118 trojan); the remaining $280 - 262 = 18$ models reconcile as 9 DQN stealth variants, 6 reward-gated-out PPO seeds, and 3 stronger adaptive DQN attacks. Of the stealth variants, one (slicing/high_load) is profiled in Table 3 and the other 8 are evaluated for ASR transfer in Section V-M, while the 6 reward-gated-out PPO seeds form the ungated-vs-gated comparison of Section V-J. Standard trojan training uses 100K fine-tuning steps with $p=10\%$ poisoning and $r_p=10$; SAC inverse_alloc uses $r_p=15$ because the target lies near a simplex corner already assigned elevated Q-value by the clean critic. PPO and A2C trojans are trained by post-hoc behavioral cloning of the actor onto the target action (10K BC steps, mean-squared-error (MSE) loss). Stealth variants use behavior-preserving regularization ($r_{\text{reg}}=5$, ratio 10%).

The defense baselines are five adapted supervised-domain methods (D1–D5: Behavioral Comparison, STRIP, Neural Cleanse, Spectral Signatures, Fine-Pruning) plus BIRD [14], the closest DRL-specific prior detector. All experiments run

TABLE 2. Model counts by DRL family. n_c : clean (calibration); n_t : trojan.

Family	n_c (seeds $\times$ tasks)	n_t	Notes
DQN	63 (21 $\times$ 3)	27 + 4 + 9 ^a	Q-gap
DDQN	21 (7 $\times$ 3)	27	Q-gap
A2C	21 (7 $\times$ 3)	27	BC on actor
SAC	21 (7 $\times$ 3)	27	Critic poisoning
PPO	24 (8 $\times$ 3) ^b	9 ^c	BC on actor

^a27 standard + 4 adaptive + 9 stealth.

^b $n_c=18$ after top-75% reward gating; see Section V-J.

^cSoftmax-normalization of $\text{Box}(0, 1)^3$ collapses the three DQN payloads to one representative continuous-action geometry.

on a single Apple M4 workstation (10 central-processing-unit (CPU) cores and 24 gigabytes (GB) of random-access memory (RAM), macOS 15.5) with Python 3.11, PyTorch 2.11, Stable-Baselines3 2.7, and Gymnasium 1.2 (CPU only, single-threaded simulator).

B. GRADFA Detection Performance

GRADFA flags all 9 standard DQN trojans and the stealth variant with no false alarms on the 21 calibration clean models (Table 3). For the focused DQN slicing profile at $n_c=21$, the calibration yields $\hat{\mu} = 2.32$, $\hat{\sigma} = 0.66$, $\tau_{99.9\%} = 4.66$; the minimum trojan ratio (12.37 $\times$) exceeds τ_α by 2.65 $\times$ and the maximum clean ratio (4.13 $\times$) sits below it with a 13% safety margin (Kolmogorov–Smirnov (KS) normality $p = 0.63$; the t -quantile $t_{20}^{-1}(0.999) = 3.55$ is 15% wider than the standard-normal $z^{-1}(0.999) = 3.09$).

The companion “DQN (pooled)” row scales the calibration to $n_c=63$ across all three tasks at 21 seeds each, with $\hat{\mu} = 2.34$, $\hat{\sigma} = 0.69$, and $\tau_{99.9\%} = 4.80$ (Student- t quantile $t_{62}^{-1}(0.999) = 3.56$). The pooled distribution preserves the separation (max clean 4.42 $\times$, $\tau/\rho_c = 1.09\times$; min trojan 10.1 $\times$, $\rho_t/\tau = 2.10\times$) with no false positives across the 63 pooled clean models, tightening the Clopper–Pearson two-sided 95% upper bound on FPR from 16.1% at $n_c=21$ to 5.7% at $n_c=63$.

The two profiles reflect a coverage variance tradeoff rather than a coverage FPR tradeoff: per-task variance dominates per-seed variance because each task’s reward landscape shapes the clean Q-distribution differently, so pooling inflates $\hat{\sigma}$ and produces a wider but still trojan-free clean envelope. Production deployments can either (a) fit the per-task t -threshold ($n_c=21$ per task) for the tightest separation, or (b) use the pooled threshold ($n_c=63$) for broader coverage at the cost of a larger margin to the worst clean observation.

Fig. 3 shows the per-group gap ratios behind this separation for one representative model in each category. For the clean model, all five feature groups produce gap ratios below 3 $\times$, indicating a uniformly flat Q-landscape. For the standard trojan, sweeping the `traffic_demand` group (which contains the `high_load` trigger) drives the gap ratio above 50 $\times$ (the trojan range across all 9 standard trojans is

TABLE 3. GRADFA detection results on the DQN slicing task with the `url1c_starve` payload ($n = 21$ clean calibration models, $\tau_{99.9\%} = 4.66$). Standard trojans cover 3 triggers with one model per trigger; the stealth row shows the representative `slicing/high_load` variant.

Model	Gap Ratio	p -value	Det?	Trigger Group
<i>Clean DQN slicing models</i> ($\hat{\mu} = 2.32$, $\hat{\sigma} = 0.66$)				
21 seeds	1.36–4.13 $\times$	0.006–0.92	0/21	—
<i>Standard trojans</i> (9 models)				
high_load ($\times 3$)	12.4–58.0 $\times$	$<10^{-10}$	3/3	traffic _{dem}
channel_drop ($\times 3$)	19.4–67.8 $\times$	$<10^{-10}$	3/3	channel _{qual}
temporal_end ($\times 3$)	28.9–46.8 $\times$	$<10^{-10}$	3/3	channel _{qual}
stealth trojan*	37.7 $\times$	$<10^{-10}$	Yes	traffic _{dem}

*Representative `slicing/high_load` stealth model; the other 8 stealth variants are evaluated for ASR transfer in Section V-M.

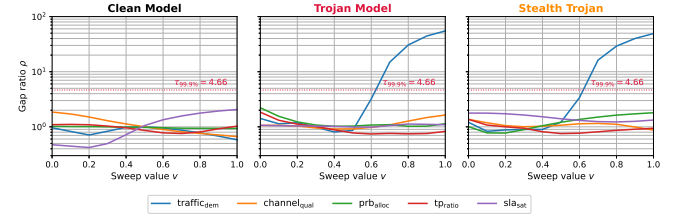


FIGURE 3. Per-feature-group Q-value gap ratios for one representative clean, standard-trojan, and stealth-trojan model. Left: clean model (all five group ratios $< 3\times$). Center: standard trojan (`traffic_demand` spikes to $\sim 55\times$). Right: stealth trojan (spike persists at $\sim 48\times$ despite behavioral regularization). Log scale; dashed line: the calibrated detection threshold $\tau_{99.9\%} = 4.66$ for the DQN slicing profile. Aggregate trojan ranges across the full set are reported in Table 3.

12.4–67.8 $\times$; Table 3), while other groups remain flat. The stealth trojan, despite achieving 73% behavioral agreement with the clean model, shows an equally pronounced spike ($\rho \approx 48\times$ on this instance, 37.7 $\times$ for the representative variant in Table 3). The Section IV prediction therefore holds: stealth regularization aligns behavior on clean inputs but cannot suppress the Q-value asymmetry at trigger states without destroying the backdoor.

C. Feature-Group Mapping Robustness

We vary only the mapping while holding 21 clean models, three trojans, 1,000 observations, an 11-point grid, and $\tau = 4.66$ fixed. “Automatic” reconstructs the declared repeated layout; “partial” retains two of three slice coordinates; “incorrect” mixes semantics; and “singleton” makes no semantic assumption. Correct, automatic, and partial mappings detect 3/3 trojans with 0/21 false positives, whereas incorrect and singleton mappings detect 1/3 (Table 4). The latter are diagnostic and cannot support a clean verdict.

TABLE 4. DQN slicing robustness to feature-group mapping quality at the unchanged threshold $\tau = 4.66$. Score ranges are clean/trojan Tier-1 Q-gap ratios.

Mapping	Score ranges	TPR	FPR
Correct semantic	[1.38, 4.45]/[53.09, 58.37]	3/3	0/21
Auto repeated-layout	[1.38, 4.45]/[53.09, 58.37]	3/3	0/21
Partial (2/3 slices)	[1.19, 2.88]/[8.67, 26.95]	3/3	0/21
Incorrect mixed semantics	[1.37, 2.68]/[2.29, 11.48]	1/3	0/21
Mapping-free singletons	[1.50, 2.75]/[1.87, 12.88]	1/3	0/21

TABLE 5. Independent clean-only calibration/test split. $n_{cal}/n_h/n_t$: calibration clean / held-out clean / test trojan. Stability is over 1,000 stratified splits; brackets give the 5th–95th percentile threshold and mean held-out FPR.

Family	$n_{cal}/n_h/n_t$	τ	TP	FP	Stability
DQN	45/15/28	4.41	28/28	0/15	[3.34, 4.41]; 1.9%
DDQN	15/6/27	3.29	27/27	1/6	[3.07, 3.76]; 6.3%
A2C	15/6/27	1.35	27/27	0/6	[0.87, 1.35]; 5.7%
SAC	15/6/27	1.55	27/27	0/6	[1.04, 1.55]; 5.5%
PPO ^a	12/6/9	2.14	9/9	0/6	[1.71, 2.14]; 8.1%
Total	102/39/118	N/A	118/118	1/39	N/A

^aTop-75%-by-reward conformance gate is applied before splitting.

D. Independent Calibration and Held-Out Validation

Table 5 reports the fixed stratified split defined in Section IV-F. Across families, clean-only calibration uses 102 models and the independent clean test uses 39; the resulting thresholds detect all 118 trojans and produce one false positive (DDQN), i.e., aggregate TPR 100% with a Clopper–Pearson 95% interval of [96.9, 100]% and FPR 2.6% with interval [0.06, 13.5]%. The wide family-level intervals, especially for $n = 6$ held-out clean models, prohibit a claim of population-level zero FPR.

Across 1,000 splits, every family retains 100% TPR and median held-out FPR of 0%, but mean FPR ranges from 1.9% to 8.1%; zero-FP splits range from 66.0% (PPO) to 76.8% (DQN). Sensitivity is stable, but the small held-out sets make FPR uncertain. Cross-task, architecture-family, seed, and environment results appear in Sections V-J and V-M.

E. Comparison with Baselines

We compare GRADFA against the five adapted supervised-domain defenses (D1–D5) and BIRD [14] on the DQN slicing calibration set of Table 3. D1–D5 and GRADFA evaluate on the 9-trojan subset (3 triggers $\times$ 3 tasks, `urllc_starve`); BIRD evaluates on a 3-trojan subset (one per trigger) because its per-target-action gradient-descent optimization takes ≈ 10 s per model. Among the methods compared (Table 6), only GRADFA achieves both 100% TPR and 0% FPR.

TABLE 6. Verification method comparison on the DQN slicing calibration set. D1–D5/GRADFA evaluate on 9 trojans; BIRD on 3 (one per trigger). Runtime is the mean over 6 runs per model.

Method	Std. TPR	Stealth	FPR	Runtime	Interp.
D1 Behavioral	9/9	No	100%	0.77 s	No
D2 STRIP	0.77 ^b	0.67 ^b	— ^c	3.20 s	No
D3 Neural Cl.	3/9	Yes*	33%	7.46 s	Partial
D4 Spectral	82%	Yes	0–53%	0.30 s	No
D5 Pruning	Collapse ^d	—	—	19.08 s	No
BIRD [14]	3/3	— ^e	30%	≈ 10 s	Partial
GRADFA	9/9	Yes	0%	0.48 s	Yes

*Detects but identifies wrong target action. ^bAUC.

^cSTRIP outputs scores; FPR depends on operator threshold.

^dPruning past 30% collapses the policy onto the trojan target.

^eBIRD trojan range [5.16, 5.36] sits inside clean [3.76, 6.47].

All methods use the same simulator, saved suspect models, CPU, and workstation (Section V-A). D1 uses 2,000 clean observations and a 50% agreement rule. D2 uses 500 clean, 500 triggered, and 1,000 reference observations, 50 blends at mixing coefficient 0.5, and a 500-threshold ROC sweep. D3 searches all 36 targets with Adam (500 steps, learning rate 0.05, $\lambda_{L1} = 0.01$, batch 128), an adaptive Q-margin, and a median-absolute-deviation threshold of 2; checks at 1,000 and 2,000 steps retain the wrong-target failure. D4 tests both layers on 900 clean/100 triggered observations and reports the higher-AUC layer at modified- $z > 2$. D5 sweeps 0–90% pruning, evaluates 30 episodes per point, and tests 50K-step clean fine-tuning. BIRD uses 200 observations, 200 Adam steps per candidate, learning rate 0.05, all 36 actions, and a clean-calibrated min-norm threshold.

We screened the recent DRL-specific alternatives discussed in Section II. SHINE requires poisoned-environment trajectories, and the activation classifier of Vyas et al. requires a trusted labeled reference set; neither input is available under our admission-time threat model. Accordingly, SHINE and the Vyas classifier are discussed qualitatively. BIRD accepts the compatible model-plus-clean-environment interface and is evaluated quantitatively.

BIRD fails for the same structural reason: it reconstructs a minimal-norm input perturbation δ such that $\arg \max_a Q(s + \delta) = a^*$ for each candidate target action a^* and uses the smallest reconstructable δ as an anomaly score [14]. The implicit assumption is that a clean model has a sharp decision boundary around the natural-action $\arg \max$, so flipping the $\arg \max$ to a non-natural action requires a large perturbation; a trojan model exposes a low-cost path through the embedded backdoor and yields a small δ . This assumption holds on Atari-class environments where action selection is high-stakes and Q-gaps are large, but it fails in our O-RAN slicing setting for the same structural reason GRADFA exploits: the clean Q-landscape is already flat ($\Delta_{\text{clean}} \approx 0.12$), so flipping the $\arg \max$ in any direction takes a small pertur-

TABLE 7. Adaptive attacker analysis: ASR vs. detectability trade-off. Tier 1/2 (ρ) are gap ratios from single-group/pairwise-group sweeps. Fixed is the legacy default ($\tau_1=3$); t -cal is the production threshold ($\tau_{99.9\%}=4.66$ from Sec. IV-F). The $r_p=5$ row evades t -cal detection while keeping 63% ASR, a residual gap (Sec. VI).

Strategy	r_p	ASR	Tier 1 ρ	Tier 2 ρ	Det? Fixed	Det? t -cal
Standard	10	100%	57.3×	61.4×	Yes	Yes
$r_p = 5$	5	63%	2.3×	2.9×	No	No
$r_p = 3$	3	5% ^a	3.4×	3.7×	Yes	No
$r_p = 2$	2	0%	1.4×	2.1×	No	No
Composite, $r_p = 10$	10	100%	31.6×	46.8×	Yes	Yes

^aBackdoor functionally inert; flagging is moot.

bation regardless of whether the model is compromised. Empirically, BIRD's trojan min-norm range [5.16, 5.36] sits in the middle of the clean range [3.76, 6.47], with no clean-trojan separation. GRADFA avoids this issue by measuring the magnitude of the Q-gap that the trigger pattern induces, rather than the magnitude of the perturbation needed to enforce a target action; the former is a direct readout of the poisoning signal, the latter is bounded above by the clean-landscape flatness.

F. Robustness Against Adaptive Attackers

An attacker aware of GRADFA could push ρ below τ_α by lowering r_p or spreading the trigger. Spreading fails: the composite variant at $r_p = 10$ keeps 100% ASR but is still flagged at Tier 2 ($\rho = 46.8\times$). Lowering r_p evades detection only by sacrificing ASR; at $r_p = 5$ the Q-gap ($\rho = 2.3\times$) hides within the clean envelope while ASR drops to 63%, and at $r_p \leq 3$ the backdoor is inert ($\text{ASR} \leq 5\%$). GRADFA flags every high-ASR ($\geq 90\%$) backdoor, but the $r_p=5$ case, a 63% intermittent attack on URLLC, remains a genuine residual gap (Section VI).

We train three detector-aware attacks with the standard 100K-step, 10%-poisoning, $r_p = 10$ budget. The first minimizes the squared top-two Q-gap on random GradFA probes every five steps (20K updates); it retains 100% ASR and clean reward 152.7 but lowers Tier 1/Tier 2 ratios to 2.98/3.91 $\times$, evading both thresholds. A three-group off-grid trigger and a sample-specific trigger also retain 100% ASR but remain detected. Direct optimization of the detector statistic can therefore evade both Q-gap tiers without reducing ASR.

The $r_p=3$ row is borderline under the fixed default $\tau_1 = 3$ but does not flag under the t -calibrated $\tau_{99.9\%} = 4.66$. The backdoor is already functionally inert at this point ($\text{ASR} = 5\%$), so this case is operationally irrelevant; flagging would require the optional Stage 1b Spectral confirmation. The GradFA-objective attack is the only tested high-ASR case that evades both Q-gap tiers. The conditional cross-

signal guard flags all 10 observation-bank repeats ($z = 14.88\text{--}16.26$) and produces 0/600 alarms in the original 60-model DQN audit. A disjoint transfer audit flags one DDQN model: 1/93 unique clean models, or 10/930 model-bank tests. The fixed guard therefore requires family-specific calibration. Randomized unequal/off-grid probing detects 0/10 attacker repeats (0/50 held-out clean alarms), and joint optimization against both signals remains open.

G. Architecture, Sequential-Trigger, and Joint-Action Validation

We test depth, temporal dependence, and joint actions outside the main 2×128 profile; these bounded cases do not establish architecture-independent coverage. On a four-layer DQN [256, 256, 128, 64], the original 4.66 threshold causes 1/12 clean alarms. Six clean seeds recalibrate Tier 1/Tier 2 to 6.77/6.75, and six disjoint seeds yield 0/6 held-out alarms. The standard recipe fails to implant a backdoor (0% ASR after 300K steps) and is excluded. A disclosed stronger recipe (20% poisoning, $r_p = 50$, 100K steps) produces six 100%-ASR trojans; all are detected (Tier 1 77.0–500.5). The six-model calibration/hold-out remains preliminary.

For memory dependence, we train six RecurrentPPO policies and implant an ordered three-event trigger in three test models. Correct-order ASR is 93.0–100%; wrong-order and last-event-only controls yield 0%, confirming order dependence. Implantation changes clean reward by -5.6% to -21.8% . A static sweep detects 0/3 trojans and flags 1/3 held-out clean models. The recurrent-state-preserving sweep tests 3,840 configurations over 128 trajectories and detects 3/3 trojans (ratios 2.22–2.56), but still flags 1/3 held-out clean models. This small study establishes temporal sensitivity, not a stable recurrent FPR.

The joint-action case couples two slicing cells and trains one centralized DQN [256, 256, 128]. Concatenated states give 30 observations; 12 local actions per cell give 144 joint actions. Ten agent-indexed groups produce 605 configurations over 512 observations in 0.66–0.72 s. Three calibration models set Tier 1/Tier 2 thresholds of 2.64/4.08; three disjoint clean models yield 0/3 alarms, and all three 100%-ASR trojans are detected. One-cell controls remain active (99.4–100% and 27.5–72.5% ASR), so the trigger is not a strict conjunction. The result covers only this centralized, reward-coupled policy, not decentralized or independently adapting agents.

H. Network Impact Assessment

GRADFA's severity stage (Stage 2, Section IV-G2) reports the network-level harm of an undetected trojan; this subsection quantifies it to show what pre-deployment detection prevents. During the `urllc_starve` trigger window (steps 50–150), PRB allocation locks to [80, 10, 10]: eMBB throughput rises 50% on the excess PRBs while the two victim slices collapse (URLLC -74% , mMTC -62% ; Table 9), and URLLC SLA satisfaction falls to 0%, a complete outage

TABLE 8. Bounded architecture, sequential-trigger, and joint-action audit. Scores are response ratios; thresholds use clean models only.

Profile / probe	n_c/n_t	Valid trojan	Clean score	Trojan score	Outcome
Deep DQN / Q-gap	12/6	6/6 at 100% ASR ^a	1.11–6.77	77.0–500.5	0/6 held-out FP; 6/6 TP ^b
Two-cell joint DQN / Q-gap	6/3	3/3 at 100% ASR	1.18–4.08 ^d	47.2–170.4 ^d	0/3 held-out FP; 3/3 TP
RecurrentPPO / static	6/3	93.0–100% ordered ASR	0.13–0.66	0.21–0.61	1/3 held-out FP; 0/3 TP ^c
RecurrentPPO / trajectory	6/3	wrong-order/last-only ASR 0%	0.16–0.70	2.22–2.56	1/3 held-out FP; 3/3 TP ^c

^aThe standard recipe failed to implant (0% ASR) and was excluded; the valid recipe uses 20% poisoning and $r_p = 50$.

^bSix clean models calibrate deep-DQN thresholds; six different clean seeds are held out.

^cThree clean models calibrate recurrent thresholds; three different clean seeds are held out.

^dRange combines Tier 1/Tier 2; three clean models calibrate and three are held out.

TABLE 9. Network impact of trojan activation (3GPP-aligned KPIs).

Metric	Slice	Pre	During	Role
Mean TP (Mbps)	eMBB	76.1	114.2	winner
	URLLC	30.7	7.9	victim
	mMTC	28.4	10.8	victim
SLA viol. rate	eMBB	8%	10%	—
	URLLC	6%	100%	critical
	mMTC	36%	99%	critical
Fairness	—	0.808	0.445	−45%

for safety-critical users. In an illustrative single-cell mix of 50 eMBB, 20 URLLC, and 200 mMTC user equipment (UEs) (loosely following 3GPP usage-class volumes [1], [31]), the victim slices hold 220 of 270 UEs, so one trigger event denies SLA to about four in five active subscribers. The exact ratio is deployment-specific, but the qualitative point holds across reasonable densities.

Fig. 4 plots the per-step network KPIs during the attack episode. During the trigger window (steps 50–150), PRB allocation locks to [80, 10, 10], causing eMBB throughput to spike while URLLC and mMTC collapse. The SLA panel confirms URLLC drops to 0% satisfaction throughout the attack, representing a complete service outage for safety-critical users. After step 150, the agent resumes normal behavior, but recovery is not instantaneous: while mMTC recovers immediately, eMBB and URLLC require 12–14 steps to re-stabilize.

I. Remediation Effectiveness

Both Stage 3 strategies suppress the tested target action (ASR $\rightarrow$ 0%) but serve different needs (Table 10). Strategy A (action masking) blocks the suspect action at trigger conditions with no retraining, suitable for emergency response. Strategy B (output-layer fine-tuning (FT)) changes the suspect model itself: in the independently regenerated repair used below, post-repair $\rho = 1.46\times$, below $\hat{\mu} = 2.32$ and $\tau_{99.9\%} = 4.66$.

ASR alone overstates recovery. On ten held-out seeds, Table 11 reports complete-episode reward and matched-window throughput, latency, SLA satisfaction, and fairness.

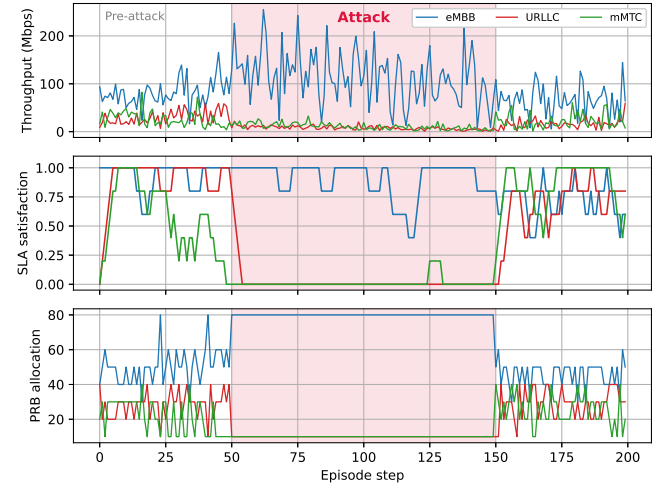


FIGURE 4. Per-step network KPIs during a trojan attack episode (high_load trigger, urllc_starve payload). Red shading: trigger active (steps 50–150). Top: per-slice throughput. Middle: SLA satisfaction (smoothed). Bottom: PRB allocation. Recovery occurs within 2–14 steps after trigger deactivation.

TABLE 10. Stage 3 remediation results on the DQN slicing/high_load/urllc_starve trojan. Clean reward uses held-out seeds 6000–6009.

Strategy	ASR	Clean Reward	Post ρ	Time
Original trojan	100%	174.3	57.3 $\times$	N/A
A: Action masking	0%	174.3	N/A ^a	instant
B: Output-layer FT	0%	157.9	1.46 $\times$	2.6 s ^b
Clean model	0%	152.2	2.78 $\times$	N/A

^aMasking is a runtime filter; model weights unchanged.

^bMean over 3 runs (range 2.4–3.0 s) on the workstation of Section V-A.

Without the trigger, masking preserves reward (174.3), while fine-tuning yields 157.9. With the trigger, both repairs keep ASR at zero, but rewards reach only 123.1 (masking) and 102.1 (fine-tuning); URLLC SLA satisfaction is 22.9% and 1.1%, respectively. Target-action suppression therefore provides only partial utility recovery.

TABLE 11. Comprehensive normal-scenario remediation evaluation (10 held-out episode seeds). Triplets are eMBB/URLLC/mMTC; window KPIs use steps 50–149. “Trig.” means the trigger remains present after mitigation.

Condition	ASR	Reward	Throughput (Mbps)	Latency (ms)	SLA sat. (%)	Fairness
Clean reference	N/A	152.2	76.9/20.4/10.9	9.09/0.43/105.02	75.4/79.5/19.5	0.605
Trojan, no trigger	N/A	174.3	59.6/20.1/21.7	11.14/0.45/37.62	64.9/83.1/82.4	0.774
Trojan, under attack	100%	92.7	108.7/6.8/8.2	6.97/2.46/114.70	90.1/0.0/5.0	0.428
Mask, no trigger	0%	174.3	59.6/20.1/21.7	11.14/0.45/37.62	64.9/83.1/82.4	0.774
Mask, trig.	0%	123.1	81.4/12.3/19.4	8.25/1.43/55.95	80.1/22.9/70.6	0.595
Fine-tune, no trigger	0%	157.9	67.2/25.4/11.7	9.94/0.38/97.54	69.9/83.6/25.2	0.685
Fine-tune, trig.	0%	102.1	83.9/7.3/22.8	8.10/2.36/28.62	84.9/1.1/85.4	0.568

Across four unseen traffic scenarios and ten seeds each, both frozen repairs keep mean and worst-case ASR at 0%. With the trigger, masking/fine-tuning obtain rewards of 94.1/90.8, URLLC SLA satisfaction of 22.1/2.0%, and fairness of 0.694/0.673. Their throughput triplets are 87.9/25.2/26.5 and 89.3/14.6/35.2 Mbps; latency triplets are 9.76/13.71/100.64 and 9.66/28.90/61.56 ms. Target-action suppression persists, but policy utility does not fully recover.

The long-horizon audit uses 2,000-step episodes with five trigger bursts: 50 episodes and 25,000 triggered steps per repair. The untreated trojan has 100% worst-burst ASR; both frozen repairs retain 0% mean, worst-episode, and worst-burst ASR. Across scenarios, reward per step ranges from 0.317–0.791 untreated, 0.349–0.902 after masking, and 0.420–0.813 after fine-tuning; trigger-window URLLC SLA ranges are 0–0.02%, 6.56–38.38%, and 0.06–3.68%. Masking leaves no-trigger reward unchanged, whereas fine-tuning changes it by -8.9% to $+55.2\%$. Because policies are frozen, this tests repeated-trigger inference stability, not online convergence.

J. Extension to Other DRL Algorithms

GRADFA generalizes from DQN to four additional algorithm families with the family-appropriate statistic from Section IV: DDQN reuses the DQN Q-gap, A2C and PPO use the action-shift statistic, and SAC uses the critic Q-gap probe. Model counts are listed in Table 2; consolidated detection results are in Table 13.

DDQN’s decoupled target update reduces overestimation bias in clean Q-values, tightening the clean baseline distribution and amplifying the trojan-induced spike: clean range $[1.20, 3.76] \times$ vs. trojan range $[13.5, 149.6] \times$, a separation wider than DQN’s. A2C shift ratios fall in $[0.00, 1.35] \times$ for clean and $[1.67, 2.79] \times$ for trojan, with a fixed $\tau = 1.5$ giving 100% TPR / 0% FPR.

PPO requires a top-75%-by-reward pre-filter before pooled separation holds. Per task, steering and power separate strictly in the ungated $n_c=8$ setting (100% TPR / 0% FPR), but 2/8 slicing seeds (bottom-quartile by clean reward) have action-shift 3.56 overlapping trojan min 3.17. Reward gating is principled because PPO is known to fail catastrophically on a fraction of seeds [36], and the functional

TABLE 12. SAC critic-gap detection robustness to candidate-action set size K . All three K values strictly separate clean from trojan and yield 100% TPR / 0% FPR.

K	Clean range	Trojan range	Sep. margin	TPR/FPR
6	[0.16, 1.76]	[2.08, 44.96]	0.32	27/27 / 0/21
12	[0.24, 1.54]	[3.87, 24.80]	2.33	27/27 / 0/21
24	[0.22, 1.54]	[3.87, 15.59]	2.33	27/27 / 0/21

Sep. margin: $\min_t \Phi_t^{\text{SAC}} - \max_c \Phi_c^{\text{SAC}}$ (absolute gap units).

conformance tests already run at onboarding (Section III-A), which are upstream of and independent of integrity screening, reject such substandard seeds before they enter the calibration set.

SAC requires critic probing because the actor-side action-shift statistic overlaps the trojan range (clean ratios reach $3.55 \times$, $\approx 92.6\%$ TPR alone). The critic Q-gap probe from Section IV-C separates strictly: clean max absolute gaps in $[0.24, 1.54]$ vs. trojan $[3.87, 24.80]$ at $K=12$. For the *inverse_alloc* payload, whose target $[0.1, 0.1, 0.8]$ lies near the corner $[0, 0, 1]$ that the clean critic already values, we train with $r_p = 15$ to keep ASR high; at the uniform $r_p = 10$, ASR on this payload drops to as low as 3% on some triggers. The higher reward does not aid detection. GRADFA still flags all nine *inverse_alloc* trojans after retraining at $r_p = 10$ ($\Phi \in [2.1, 19.7]$ vs. clean ≤ 1.54), so the SAC result is not an artifact of the higher poison reward.

The choice $K=12$ does not drive the result. Re-running the critic-gap evaluation at $K \in \{6, 12, 24\}$ on the same 21 clean and 27 trojan SAC models yields 100% TPR / 0% FPR at all three points (Table 12); the discriminative signal is already concentrated in the trojan-target geometry covered by $K=6$, and adding more candidates only confirms the ordering.

Table 13 consolidates results across the 5 families. All five reach 100% TPR / 0% FPR on the focused per-task profile (102 clean, 99 trojan, PPO on its reward-gated $n_c=18$ set), and the optional pooled-DQN profile extends this to 144 clean and 118 trojan models while preserving 0% FPR. The per-family statistic (Q-gap for DQN/DDQN, action-shift for PPO/A2C, critic Q-gap for SAC) reflects where the

TABLE 13. GRADFA detection across 5 DRL algorithm families. n_c/n_t : clean / trojan model counts. DQN and DDQN use t -calibrated thresholds at $n_c \geq 21$; A2C, SAC, PPO use fixed-scalar thresholds (PPO at $n_c=18$ top-75% reward-gated of 24). A representative DQN stealth variant is profiled separately in Table 3; the remaining 8 stealth variants are evaluated for ASR transfer in Section V-M.

Algo	n_c/n_t	Trojan range	TPR	FPR
DQN (slicing)	21/9	12.4–67.8×	100%	0%
DQN (pooled)	63/28 [‡]	10.1–85.0×	100%	0%
DDQN	21/27	13.5–149.6×	100%	0%
A2C	21/27	1.7–2.8×	100%	0%
SAC	21/27	3.9–24.8	100%	0%
PPO [†]	18/9	2.5–3.6×	100%	0%
Focused (5 fam.)	102/99	—	100%	0%
+ Pooled DQN[‡]	63/28	—	100%	0/63

Statistic: Q-gap for DQN/DDQN, action-shift for A2C/PPO, critic Q-gap for SAC ($K=12$, K-ablation in Table 12).

[†]PPO uses 1 representative payload (action-geometry).

[‡] $n_t = 27$ standard +1 adaptive composite trojan.

[‡]Pools the 3 task-specific calibration sets (21 seeds each) into a single threshold; Clopper–Pearson 95% FPR upper bound: 5.7% at $n_c=63$ (versus 16.1% at $n_c=21$).

attacker injects backdoor signal in each architecture, but the underlying detection principle of feature-group sweep with anomaly thresholding is shared.

K. Multi-xApp Redundancy as Defense-in-Depth

GRADFA operates pre-deployment; for trojans that evade screening, runtime redundancy across multiple xApps from different vendors provides a complementary post-deployment defense-in-depth layer that production Near-RT RICs already support. Table 14 evaluates whether multi-xApp conflict resolution can neutralize a trojan even if it evades pre-deployment screening. With clean majority (1T+2C), all three strategies achieve 0% effective ASR. Weighted average and consensus strategies also neutralize trojans under attacker majority (2T+1C), because averaging dilutes the extreme allocation and consensus rejects it as an outlier. Multi-xApp deployment thus adds a second layer that complements pre-deployment verification.

L. Computational Overhead

Per-model wall-clock runtimes, each the mean of 6 runs on the workstation of Section V-A, are reported in Table 6 alongside detection accuracy. GRADFA at 0.48 s is on par with the cheapest baselines (D1 at 0.77 s, D4 at 0.30 s) and roughly 15× faster than Neural Cleanse, the closest baseline in detection accuracy. The 0.48 s breaks down approximately as ≈ 0.36 s for the $N=1000$ observation rollout and ≈ 0.12 s for the two-tier sweep (165 batched forward passes over the rollout); reusing the rollout across submissions cuts the per-model cost to ≈ 0.12 s. On a value-based model that clears

TABLE 14. Multi-xApp conflict resolution vs. trojan effectiveness. Configurations are written as $nT+nC$ (trojan/clean xApps in deployment). Effective ASR is the fraction of triggered steps where the resolved action equals the trojan target; Override Rate is the fraction of triggered steps where the resolution mechanism overrides the trojan's proposed action.

Config	Strategy	Eff. ASR	Override Rate	SLA
1T only	—	100%	0%	53%
1T+2C	Majority	0%	100%	60%
	Wtd. Avg	0%	100%	66%
	Consensus	0%	100%	68%
2T+1C	Majority	100%	0%	53%
	Wtd. Avg	0%	100%	61%
	Consensus	0%	100%	61%

both tiers, the batched cross-signal guard adds a 10-run mean of 0.038 s excluding the shared rollout, for an end-to-end screening time of approximately 0.52 s. GRADFA attains this speedup by replacing the per-target, iterative trigger reconstruction of Neural Cleanse and BIRD with a fixed sweep and conditional cross-check, lowering the cost by an order of magnitude. GRADFA runs once at admission, off the 10 ms–1 s RIC control path, so the overhead is borne by the marketplace onboarding workflow rather than the runtime control loop.

M. Cross-Environment Transferability

We evaluate whether trojan effectiveness obtained in our simulator transfers to other network environments. All 18 trojan models (9 standard + 9 stealth) are tested across six environments spanning synthetic, 3GPP-calibrated, and real-world traces: the training simulator (*Synthetic*), three 3GPP-calibrated channel models (*Urban Macro*, *Urban Micro*, *Rural*), a Colosseum testbed trace (*CoLo-RAN*), and a real 5G network trace (*TRACTOR*).

Table 15 (top block) summarizes the attack-transfer results. Standard trojans transfer at 100% ASR across all environments for 8 of 9 configurations; the sole outlier (power/channel_drop) drops to 57–62% on 3GPP-calibrated traces but recovers to 100% on real traces (CoLo-RAN, TRACTOR). Stealth trojans transfer with higher consistency, with mean ASR $\geq 97.5\%$ across all environments.

Trojans trained on synthetic data transfer better to real-world traces (CoLo-RAN: 100%, TRACTOR: 100%) than to 3GPP-calibrated traces (95.6%), suggesting that calibrated simulators introduce artificial distribution shifts absent in real networks.

Detection-side transferability. The attack-transfer numbers above only confirm that the trojan policy fires when the trigger is presented; the marketplace use case additionally requires that the GRADFA detector itself works when the operator hosts the suspect xApp in a different observation

TABLE 15. Cross-environment transferability across the six observation environments. *Top*: mean attack ASR (%) over the 18 trojan models (9 standard + 9 stealth). *Bottom*: GRADFA detection TPR / FPR (%) per algorithm after re-collecting the $N=1000$ baseline rollout from each environment and applying the family-specific calibration of Section IV-F; PPO[†] uses a synthetic-anchored rollout (Section V-M).

	n	Synth.	3GPP ^a	CoIo	TRAC
<i>Attack: mean ASR (%)</i>					
Standard ^b	9	100	95.6	100	100
Stealth	9	97.6	98.0	98.9	99.9
<i>Detection: TPR / FPR (%), $n = n_c/n_t$</i>					
DQN	63/28	100/0	85.7/0	100/0	100/0
DDQN	21/27	100/0	81.5/0	100/0	92.6/0
A2C	21/27	100/0	100/0	96.3/0	96.3/0
SAC	21/27	100/0	100/0	100/0	100/0
PPO [†]	18/9	100/0	100/0	100/0	100/0

^aAttack: mean over Urban Macro / Micro / Rural;

Detection: worst-of-three across the same three.

^bStandard pw/ch_dr drops to 60.7% on 3GPP; other 8 standard configs = 100%.

[†]Synthetic-anchored baseline rollout.

environment from the one we used for calibration. We therefore re-run the full GRADFA pipeline (per-family statistic, per-family threshold; Section IV) on the pooled per-family calibration set (DQN: 63 clean / 28 trojans across the three tasks at 21 seeds each; DDQN, A2C, SAC: 21/27; PPO: 18/9 after reward gating), but now collecting the $N=1000$ baseline rollout from each of the six environments above. Table 15 (bottom block) reports the resulting TPR/FPR per family per environment.

The outcomes split into two regimes. SAC and PPO hold 100% TPR / 0% FPR across all six environments; A2C drops to 96.3% on CoIo-RAN/TRACTOR (one trojan). DQN/DDQN are weakest under environment shift: DQN drops to 85.7% on the worst 3GPP channel (recovers to 100% on real traces); DDQN drops to 81.5% on 3GPP and 92.6% on TRACTOR. FPR stays at 0% in every cell, so misses are false negatives, not false alarms. All DDQN misses are *high_load* trojans, the weakest trigger; their ratios (5.1–6.6 $\times$) clear the clean envelope ($\leq 4.10\times$) but fall under a $\tau_{99.9\%}$ inflated by the 7-model slicing calibration, so a 99% level recovers most at 0% FPR. The degradation tracks the attack-transfer artifact in the top block: calibrated channels shift the baseline rollout away from the training distribution, weakening the trigger Q-gap ratio. A deployment-time fix is to recalibrate the per-task threshold on the operator's own E2 trace pool, which the screening pipeline of Section IV-G1 supports as a one-time onboarding step (Section VI). Overall, across all five families and six environments, detection holds at 100% TPR / 0% FPR under per-task calibration and drops to $\geq 81.5\%$ only under foreign 3GPP channels, recoverable by recalibration.

VI. Discussion

A. Deployment Implications

The full pipeline completes in about 3 seconds per submission (screening 0.48 s, or 0.52 s with the conditional guard; severity 0.05 s; optional remediation 2.6 s), compatible with xApp marketplace onboarding timelines where model certification typically takes hours to days. The severity report distinguishes high-severity trojans targeting URLLC, which warrant immediate rejection, from low-severity detections routed to quarantine and manual review.

B. Limitations and Failure Modes

GRADFA is an admission-time empirical screen for policy artifacts with a verifiable observation schema. The following boundaries identify when it fails or may fail; a deployment outside them must not inherit the reported TPR/FPR.

- **Feature semantics are a trust boundary.** Correct, automatic, and partial mappings detect 3/3 tested trojans, whereas incorrect and singleton mappings detect 1/3. Proprietary embeddings or undocumented preprocessing require quarantine or a validated adapter; singleton sweeps are diagnostic only.
- **A statistic-aware attacker can evade a component.** The GradFA-objective attack retains 100% ASR while evading both Q-gap tiers. The guard catches 10/10 repeats but flags one DDQN model in transfer (1/93 models; 10/930 model-bank tests), requiring family-specific calibration. Joint optimization against both signals, dynamic rewards, and cascading successor-state attacks remain open.
- **Threshold guarantees are finite-sample and deployment-specific.** The independent split detects 118/118 trojans with 1/39 held-out clean alarms (95% FPR interval [0.06, 13.5]%; mean FPR over repeated splits is 1.9–8.1%. Larger deployment-specific clean pools are required. Proposition 1 and Lemma 1 are conditional, not out-of-sample guarantees.
- **Architecture and temporal coverage are preliminary.** Architecture-specific calibration detects 6/6 deep-DQN trojans with 0/6 held-out alarms. The recurrent sweep detects 3/3 but flags 1/3 held-out clean models; the centralized two-cell audit detects 3/3 with 0/3 alarms, but its trigger is not a strict two-cell conjunction. Graph, transformer, decentralized, independently learning, and larger multi-agent policies remain untested.
- **Remediation may not recover service.** Both repairs retain 0% ASR over 25,000 triggered steps, yet trigger-window URLLC SLA remains 6.56–38.38% for masking and 0.06–3.68% for fine-tuning. Frozen-policy evaluation does not establish online convergence; every repair requires KPI validation or rejection/rollback.
- **Live adaptive deployment remains untested.** Environment transfer uses trace replay, the two-cell policy is centralized, and the multi-xApp policies are frozen.

Shared-state feedback, peer adaptation, and coordinated multi-agent backdoors may alter detection and remediation.

VII. Conclusion

We have proposed GRADFA, a pre-deployment backdoor detection framework for DRL-based xApps, and embedded it in an end-to-end pipeline that adds severity assessment and remediation. GRADFA exploits the asymmetry between legitimate and compromised policy responses to systematic feature sweeps, instantiated as a Q-gap statistic for DQN/DDQN, an action-shift statistic for PPO/A2C, and a critic Q-gap statistic for SAC. On the focused per-task profile (102 clean / 99 trojan models across five DRL algorithm families), GRADFA reaches 100% TPR at 0% FPR; a pooled DQN configuration further extends coverage to 144 clean and 118 trojan models while maintaining 0% FPR (Clopper–Pearson 95% upper bound 5.7% at $n_c=63$). In the independent clean-only split, it detects 118/118 trojans with 1/39 held-out clean false positives (TPR 95% interval [96.9, 100]%; FPR interval [0.06, 13.5]%), making the empirical scope explicit. Under foreign 3GPP-channel rollouts the value-based statistic drops to $\geq 81.5\%$ TPR (0% FPR) and is restored by operator-side recalibration.

Beyond detection, the pipeline provides network-impact-aware severity assessment (URLLC throughput -74% ; 220/270 UEs across victim slices affected) and targeted remediation ($ASR \rightarrow 0\%$, $\rho : 57.3\times \rightarrow 1.46\times$). Both repairs retain 0% worst-burst ASR but recover utility only partially. The GradFA-objective attacker evades both Q-gap tiers at 100% ASR; the guard catches 10/10 repeats but flags 1/93 clean models in transfer, so family-specific calibration and joint-signal evasion remain open. Deep, recurrent, and centralized two-cell audits provide bounded positive evidence, not graph, transformer, or decentralized multi-agent coverage. Mapping errors also cause misses. Multi-xApp redundancy with conflict resolution provides an orthogonal post-deployment layer: weighted-averaging and consensus strategies neutralize the trojan even under attacker majority, while plain majority voting fails. The full pipeline (screening, severity assessment, and optional remediation) completes in about 3 seconds per submission, fitting within typical xApp marketplace onboarding timelines.

Appendix A

Proof of Proposition 1

By the Bellman fixed point (A3), $Q_\theta(s_{\text{trig}}, a) = \hat{r}(s_{\text{trig}}, a) + \gamma U(a)$ for every action a , where $U(a) := \mathbb{E}_{s' \sim \hat{P}(\cdot | s_{\text{trig}}, a)}[V_\theta(s')]$ is the continuation value from (A4).

Target action. By the conditional-poisoning assumption (A2), a fraction α of the transitions at (s_{trig}, a^*) carry reward r_p and the remaining $(1-\alpha)$ are clean with $|r| \leq r_c^{\text{sup}}$, so the empirical mean reward obeys $\hat{r}(s_{\text{trig}}, a^*) \geq \alpha r_p -$

$(1-\alpha) r_c^{\text{sup}}$. Hence

$$Q_\theta(s_{\text{trig}}, a^*) \geq \alpha r_p - (1-\alpha) r_c^{\text{sup}} + \gamma U(a^*). \quad (20)$$

Non-target actions. For any $a \neq a^*$ all transitions are clean by (A2), so $\hat{r}(s_{\text{trig}}, a) \leq r_c^{\text{sup}}$ and

$$Q_\theta(s_{\text{trig}}, a) \leq r_c^{\text{sup}} + \gamma U(a). \quad (21)$$

Gap lower bound. The top-two gap satisfies $\Delta(s_{\text{trig}}) = \max_a Q_\theta - Q_\theta^{(2)} \geq Q_\theta(s_{\text{trig}}, a^*) - \max_{a \neq a^*} Q_\theta(s_{\text{trig}}, a)$ for any distinguished action a^* (equality when a^* is the maximizer; otherwise the left side is nonnegative while the right side is nonpositive). Substituting (20) and (21),

$$\Delta(s_{\text{trig}}) \geq \alpha r_p - (2-\alpha) r_c^{\text{sup}} + \gamma \left[U(a^*) - \max_{a \neq a^*} U(a) \right].$$

The bounded-continuation-gap assumption (A4) gives $\max_{a \neq a^*} \max(U(a) - U(a^*), 0) \leq C \Delta_c^{\text{max}}$, hence $U(a^*) - \max_{a \neq a^*} U(a) \geq -C \Delta_c^{\text{max}}$ and $\gamma[\cdot] \geq -C\gamma \Delta_c^{\text{max}}$. Therefore

$$\Delta(s_{\text{trig}}) \geq \alpha r_p - (2-\alpha) r_c^{\text{sup}} - C\gamma \Delta_c^{\text{max}} = L_\alpha.$$

Since $L_\alpha > 0$ by hypothesis, $\Delta(s_{\text{trig}}) > 0$, and dividing by $\Delta_{\text{baseline}} \leq \Delta_c^{\text{max}}$ from (A1) yields $\rho_{\text{tab}}(s_{\text{trig}}) = \Delta(s_{\text{trig}})/\Delta_{\text{baseline}} \geq L_\alpha/\Delta_c^{\text{max}}$. Setting $\alpha = 1$ recovers $L_\alpha = r_p - r_c^{\text{sup}} - C\gamma \Delta_c^{\text{max}}$. $\square$

Appendix B

Proof of Lemma 1

For any $\theta_t \in \mathcal{T}_F^{\text{cal}}$, restricting the maximum in (12) to the pair (g^*, v^*) guaranteed by (B1) and dividing by $Z_F(\theta_t) > 0$ gives $\rho_F(\theta_t) \geq \kappa_F(\theta_t) M_F(\theta_t)/Z_F(\theta_t) \geq L_F$. For any $\theta_c \in \mathcal{C}_F^{\text{cal}}$, $\rho_F(\theta_c) \leq \rho_F^{\text{clean, max}}$ by definition. Under (B2), any τ_F in the stated interval therefore satisfies $\rho_F(\theta_c) \leq \rho_F^{\text{clean, max}} < \tau_F < L_F \leq \rho_F(\theta_t)$, which is (16). $\square$

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